Brighter Blue

Core Testing>Basic Testing>Day 1

Student Guide

IBM

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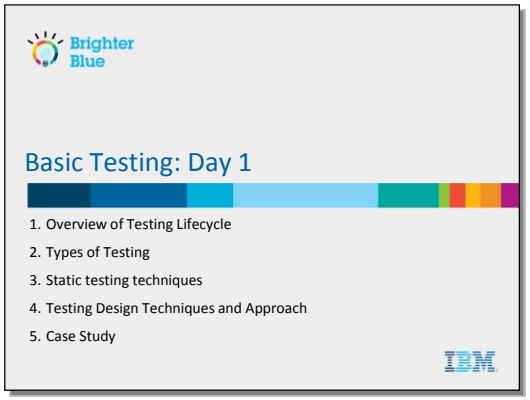
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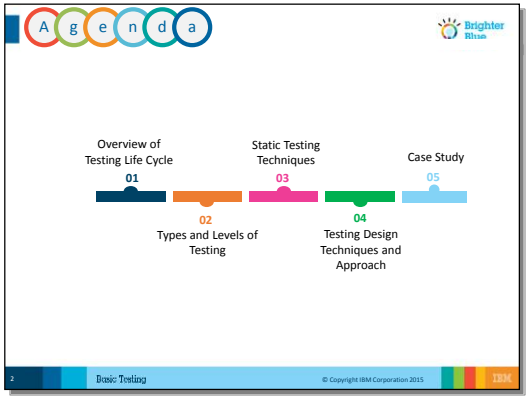
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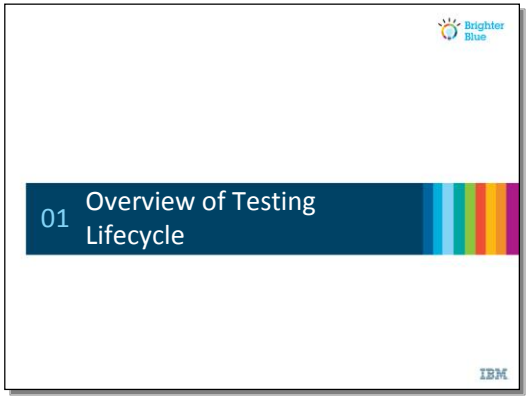
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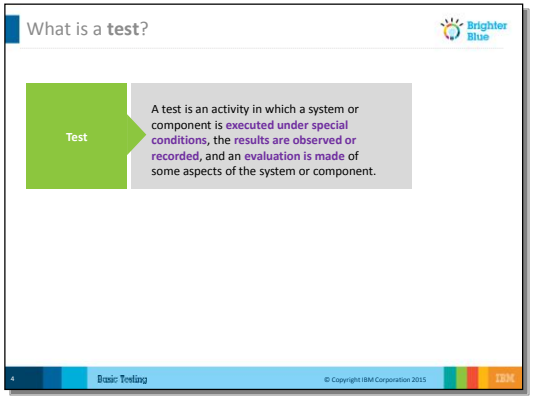
Module 01: Overview of Testing Life Cycle

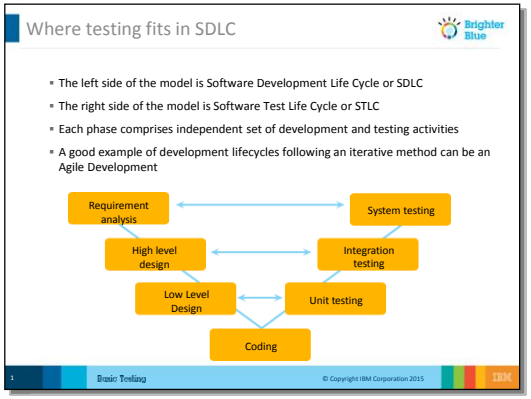
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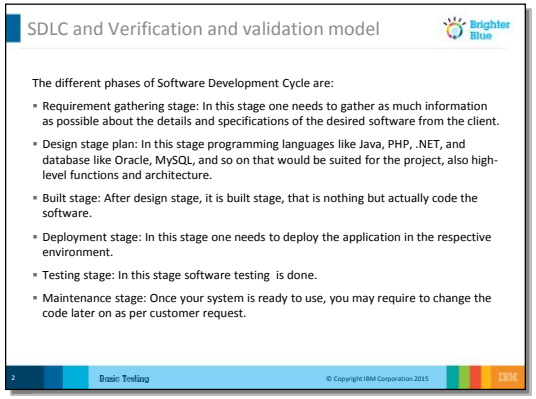
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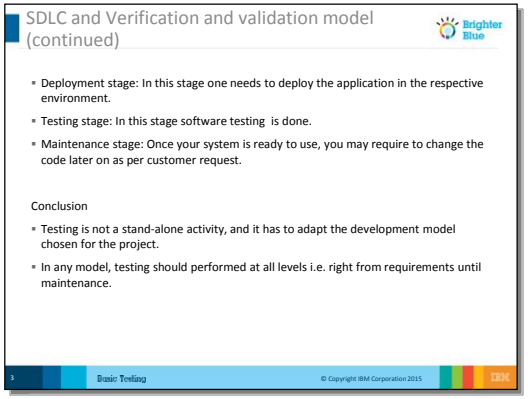
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Slide 3  **At the end of this module, you should be able to:** • **Give an overview of testing lifecycle** • **Describe the various test strategies** • **Describe the varied facets of test planning** • **Recall the process of Test Case creation** • **Recall the process of test execution** • **List the testing types** • **Describe default workflow**	

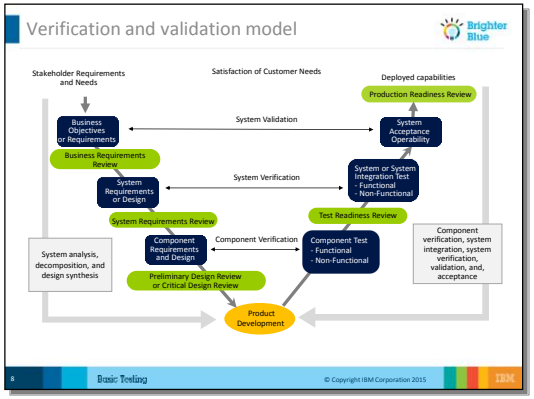
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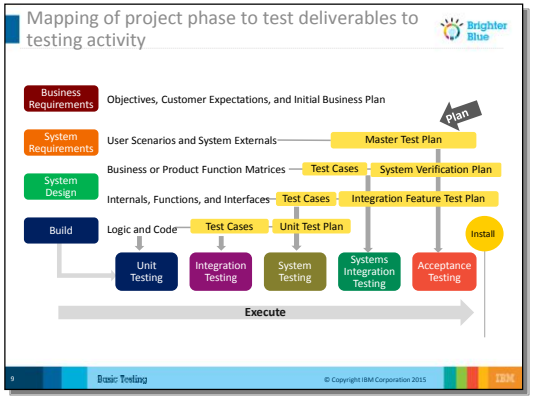
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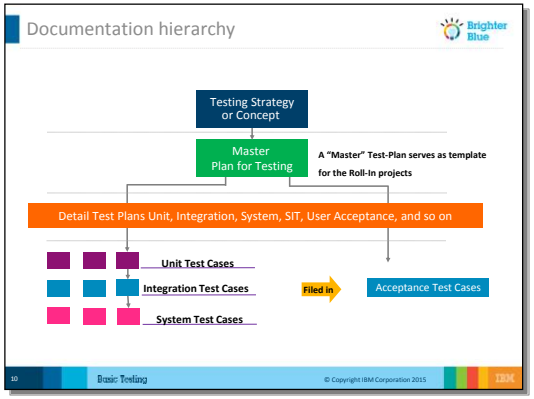
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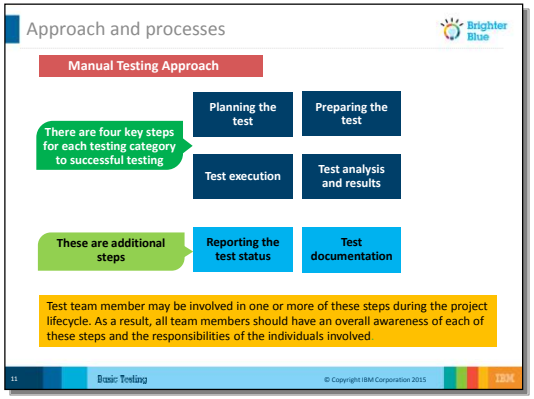
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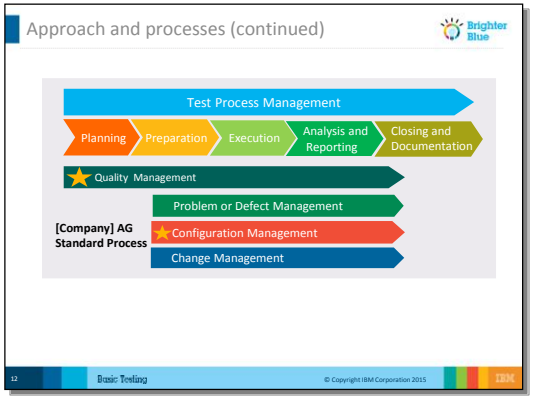
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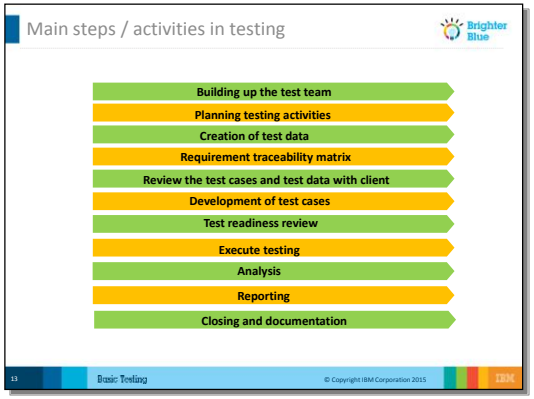
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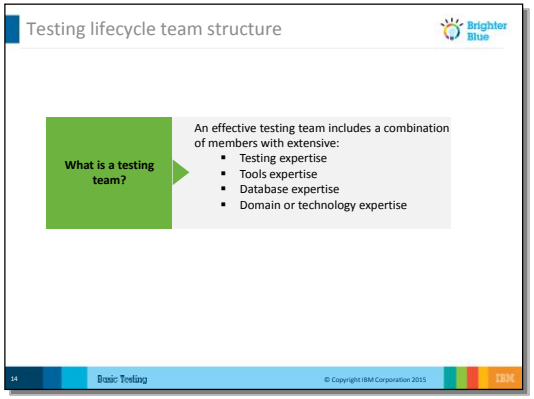
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Slide 13  **Building up the test team:** • **Recruit and Onboard team members** • **Assign team members to roles (Business and IT)** • **Coach testing team members (Key User on Application** **Planning testing activities:** • **Analyze business requirements** • **Estimate test effort**	

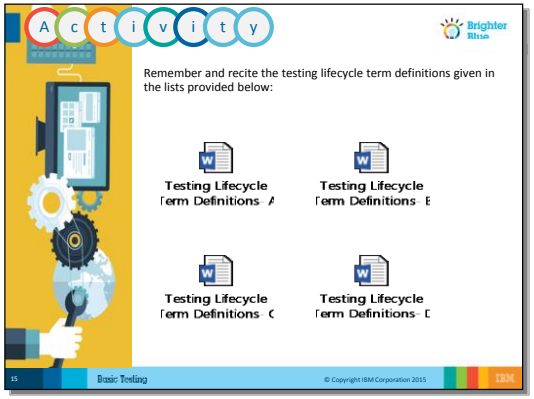
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• Create Project Plan • Prepare test plan or test strategy document for various types of testing • Select test tool • Plan resource and determine roles and responsibilities Creation of test data: • Align with business user and business analyst to define demand for test data • Create test data Requirement traceability matrix: • Prepare the requirement traceability verification matrix or RTVM by mapping each requirement with test cases Review the test cases and test data with client: • Send the initial version of test cases for review • Close all the review comments by updating the test cases • Get signed off test cases and test data	

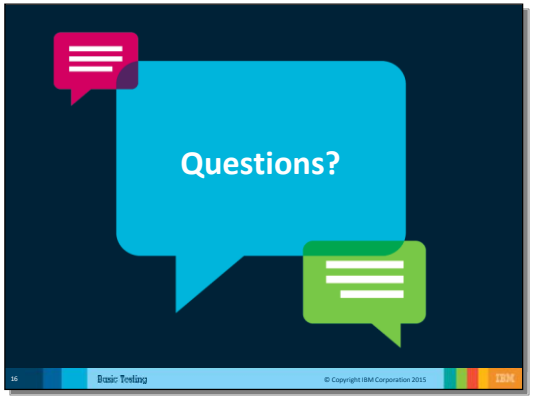
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• Archive approved test cases and test data Development of test cases: • Identify test cases • Evaluate test cases regarding risk categories • Define test entry and exit criteria • Describe test cases • Peer review of test cases Test Readiness Review • Verify test environment is setup and stable for execution • Verify unit test results pass the entry criteria of testing phase • Client-approved test cases and test data • Perform Smoke test Execute Testing • Execute tests as per plan	

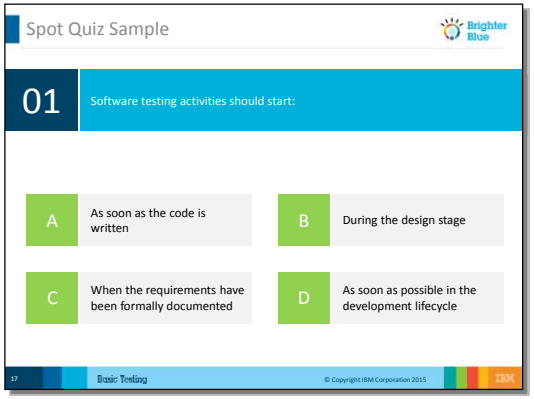
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• Document test results, and log defects for failed cases • Update test plans or test cases, if necessary • Retest the defect fixes and tack the defects to closure • Do regression testing of application **Analysis** • Analyze the Test results • Prepare Test Metrics:Time, Test coverage, Cost, Software Quality, and Critical Business Objectives • Perform Causal analysis– prepare Causal Analysis Report **Reporting** • Publish Test Metrics • Publish Test reports: Test Summary report, Causal Analysis Report, and Audit Report **Closing and Documentation** • Analyze the Test results	

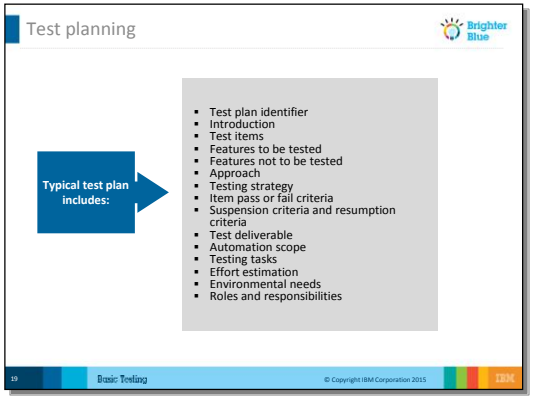
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• Prepare Test Metrics- Time, Test coverage, Cost, Software Quality, and Critical Business Objectives • Perform Causal analysis– Prepare Causal Analysis Report	

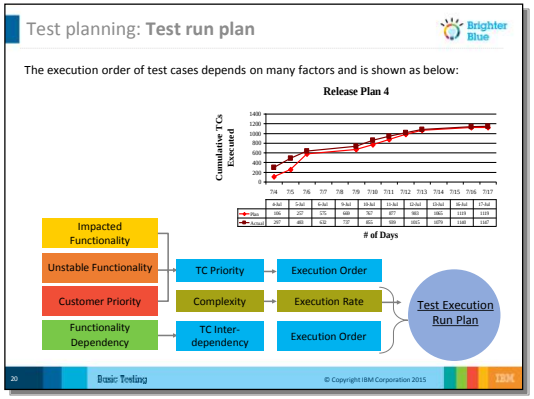
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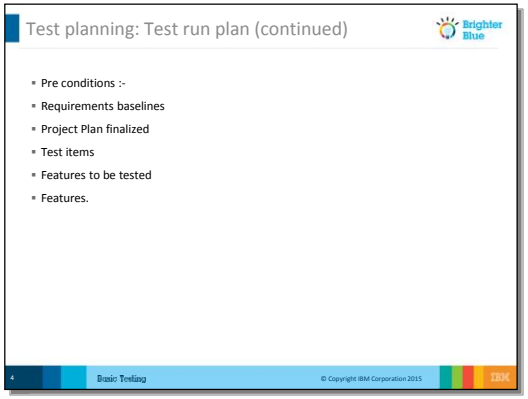
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Slide 15  **Refer to the Supporting Documents_Day 1 folder for the embedded files.**	

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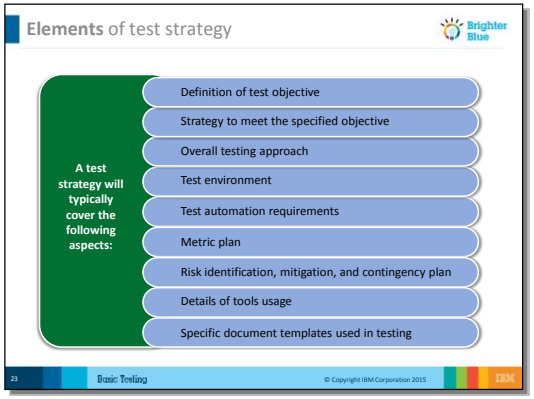
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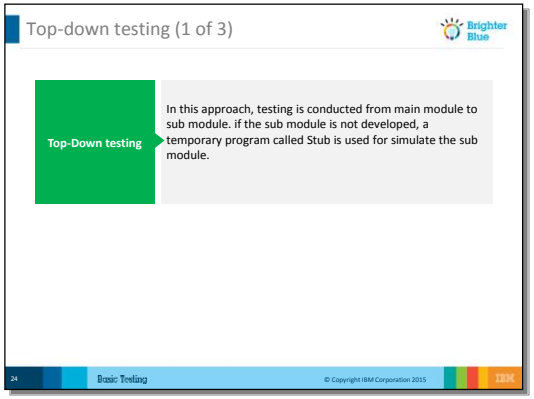
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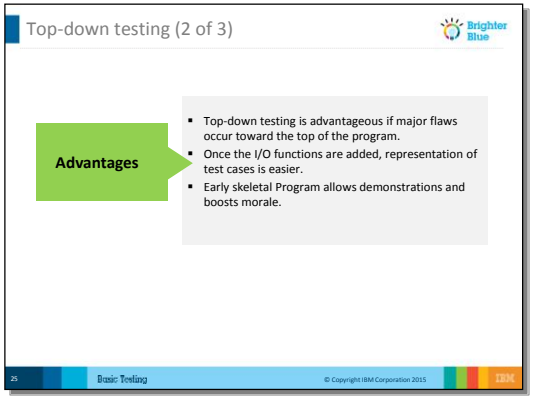
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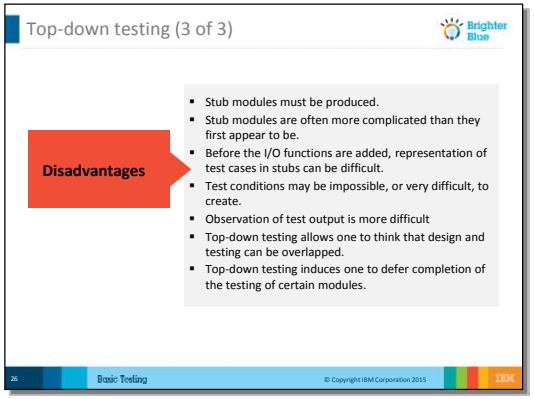
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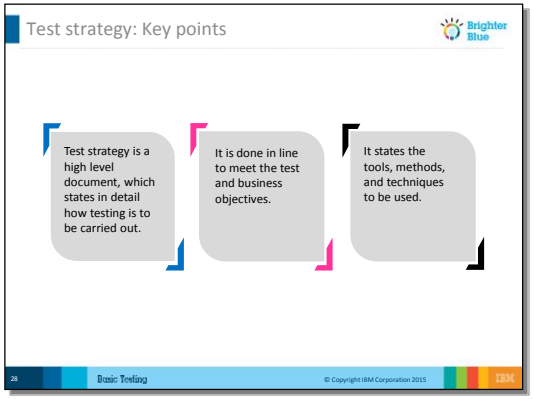
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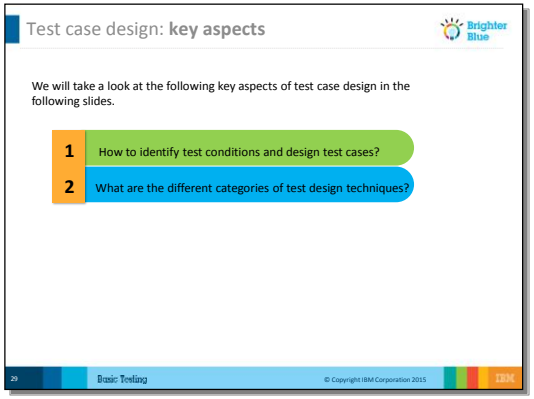
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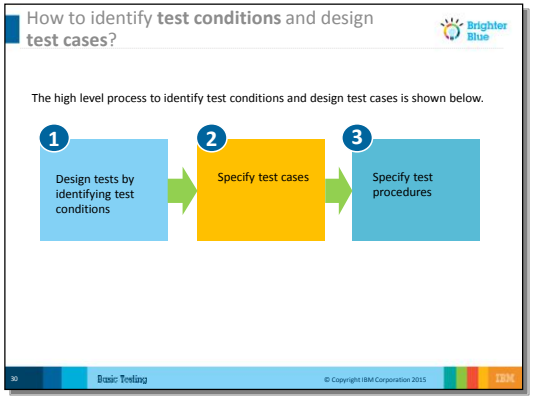
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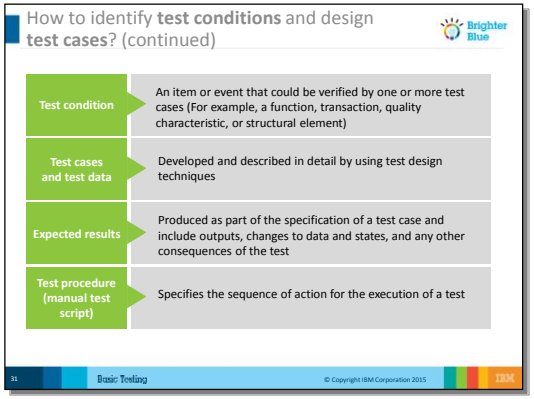
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Slide 25  Example: Suppose Sales Order Printing program uses another unit which calculates Sales discounts by some complex calculations. Then call to this unit will be replaced by a Stub, which will simply return fix discount data.	

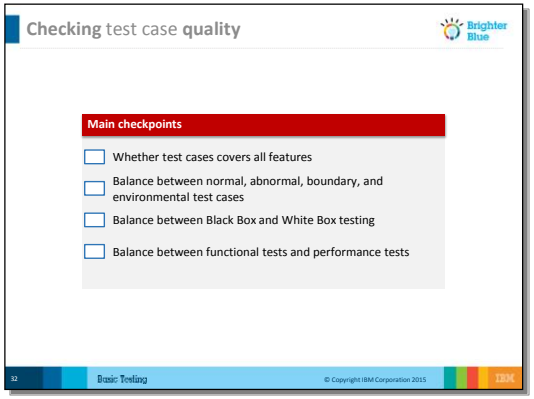
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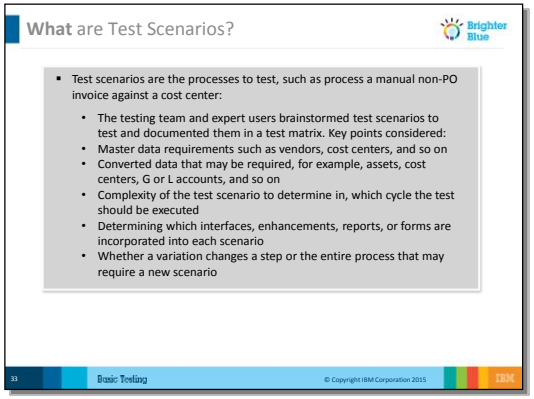
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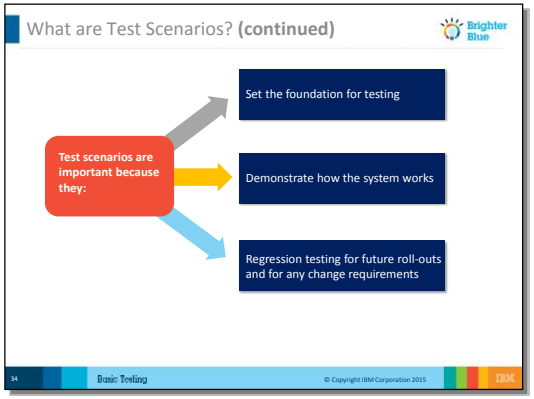
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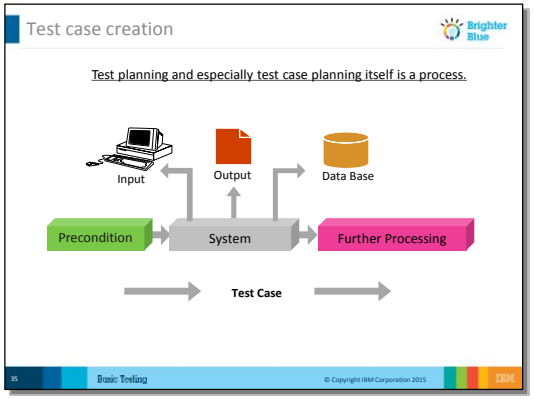
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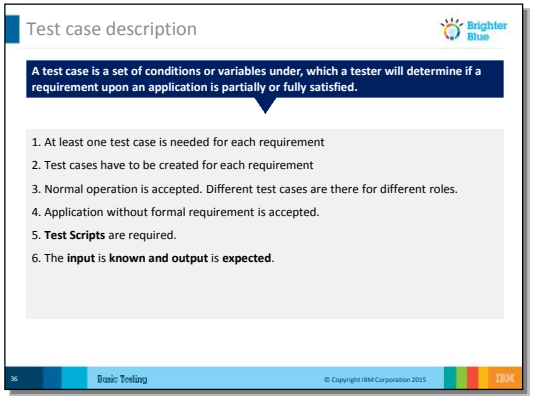
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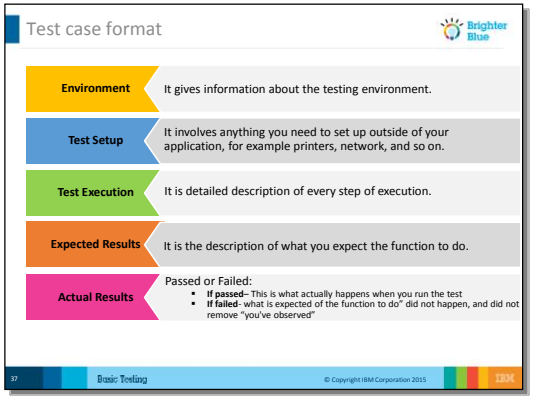
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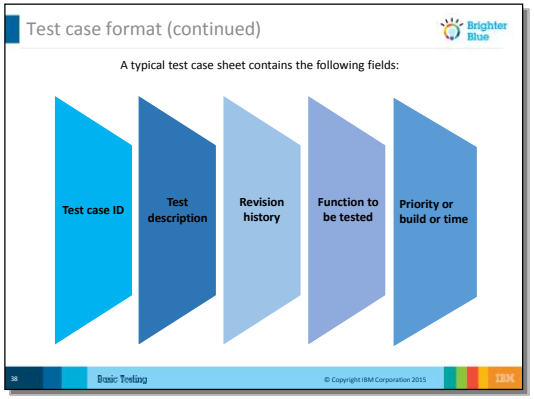
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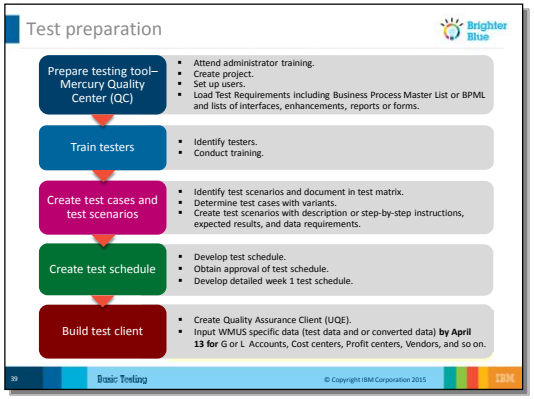
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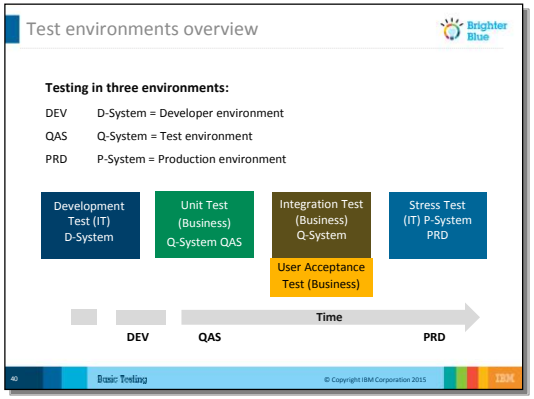
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Slide 35  At least one test case for each requirement - It may take many test cases to determine that a requirement is fully satisfied. In order to fully test that all the requirements of an application are met, there must be at least one test case for each requirement unless a requirement has sub requirements. Create test cases for each requirement - In that situation, each sub requirement must have at least one test case, or we should be creating at least two test cases for each requirement. One of them should perform positive testing of requirement and other should perform negative testing. Create Test cases for different roles (for example, manager view, user view, group, and so on.	

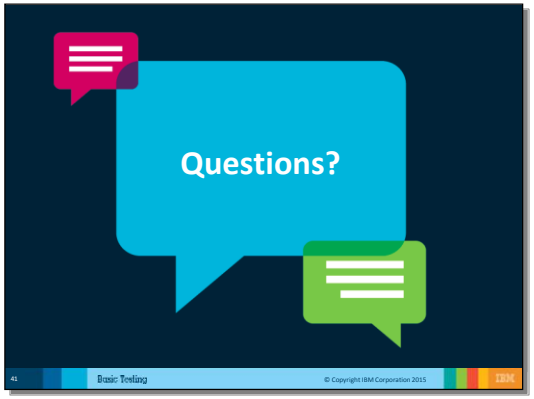
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Accepted normal operation - If the application is created without formal requirements, then test cases are written based on the accepted normal operation of programs of a similar class. Test cases for different roles - Create Test cases for different roles (for example, manager view, user view, group, and so on. Application without formal requirement - If the application is created without formal requirements, then test cases are written based on the accepted normal operation of programs of a similar class. Test Scripts - A test script is the executable form of a test. Known input and expected output - What characterizes a formal, written test case is that there is a known input and an expected output, which is worked out before the test is executed. The known input should test a precondition and the expected output should test a post condition.	

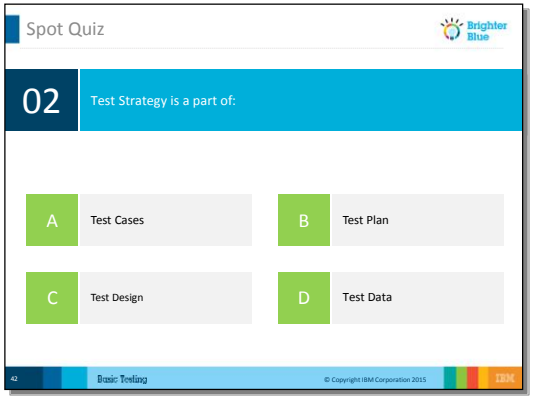
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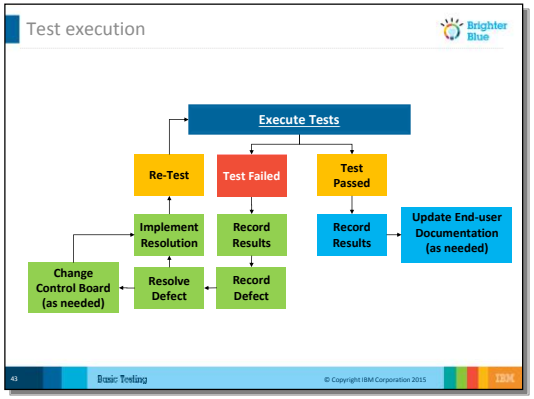
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Slide 37  A typical test case sheet contains the following fields … Test case ID: It is a unique number given to test case in order to be identified. Test description: This is the description of test case you are going to test. Revision history: Each test case has to have its revision history in order to know when and by whom it is created or modified. Function to be tested: This is the name of function to be tested. Priority or build or time: The priority of the test case, build version of the test case and the minimum time-to-execute the test case are also mentioned.	

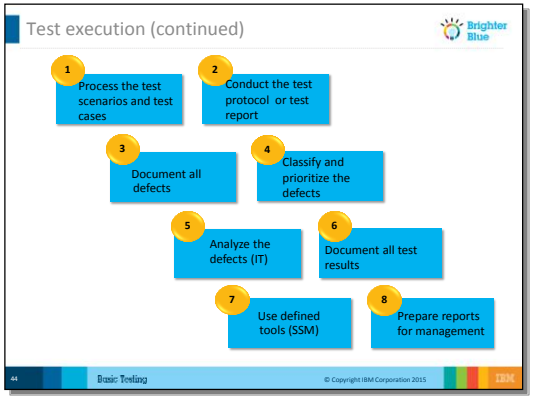
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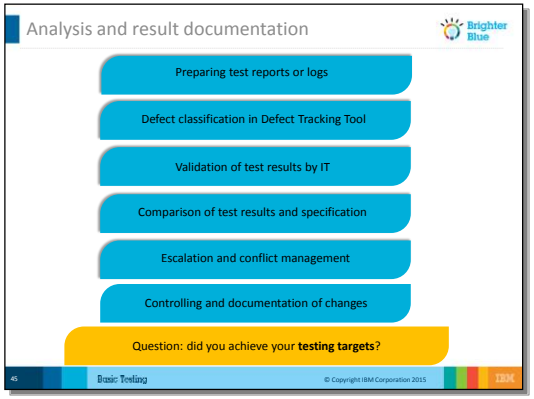
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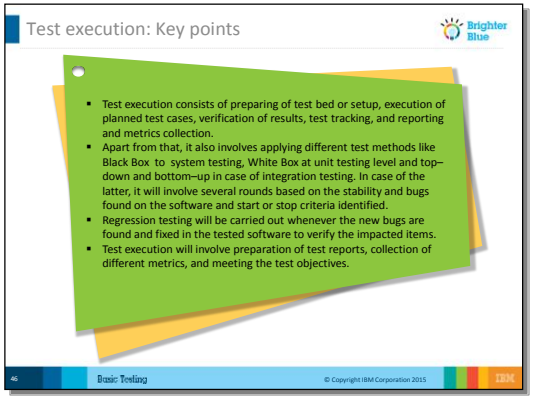
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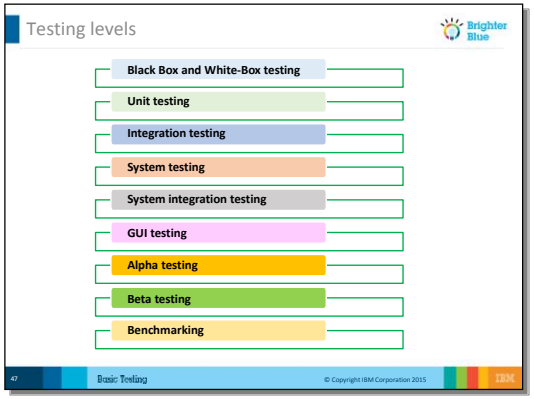
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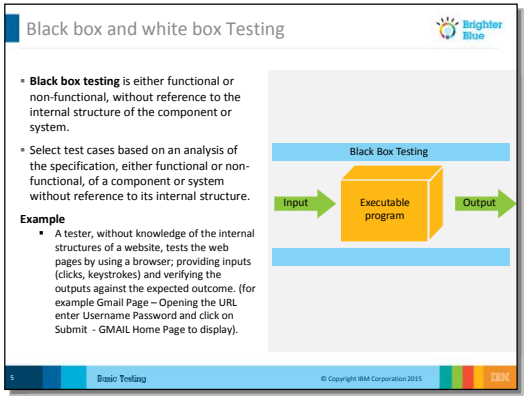
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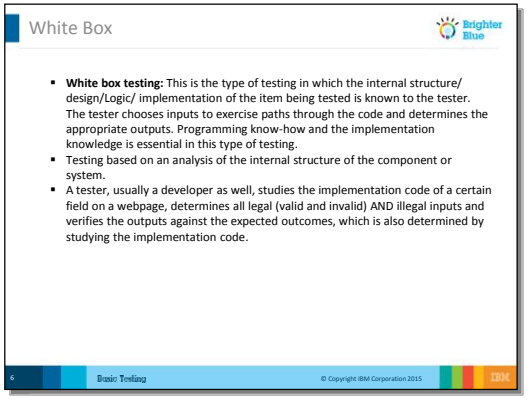
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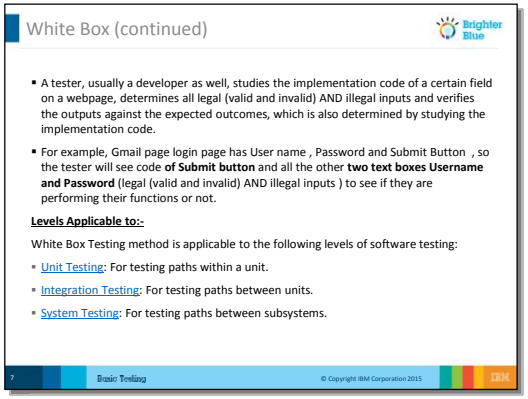
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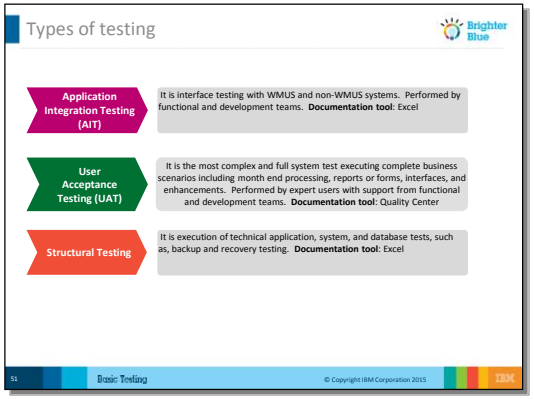
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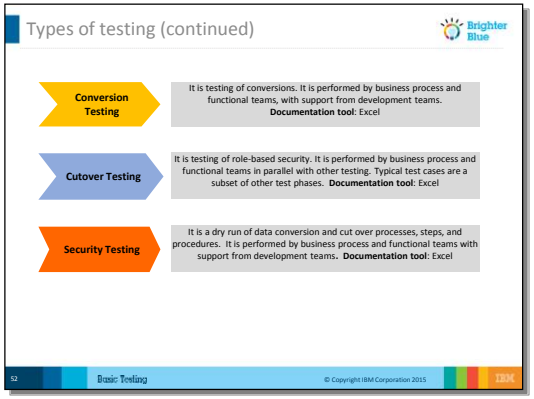
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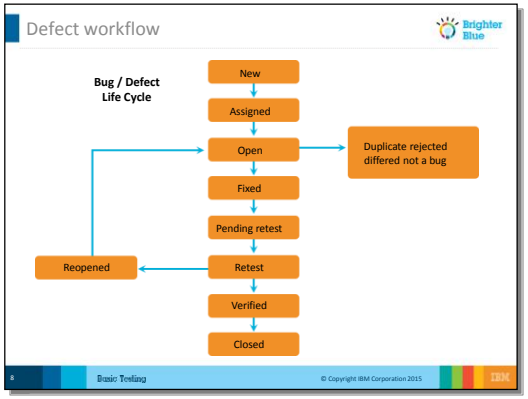
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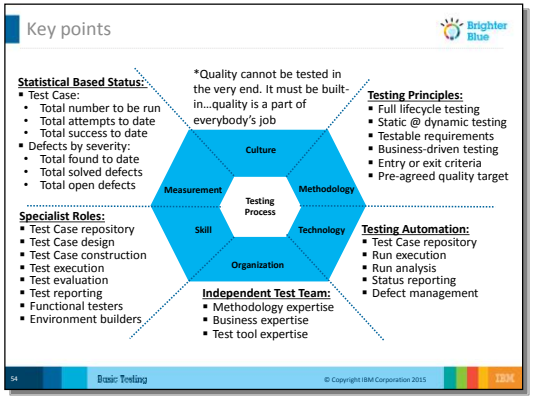
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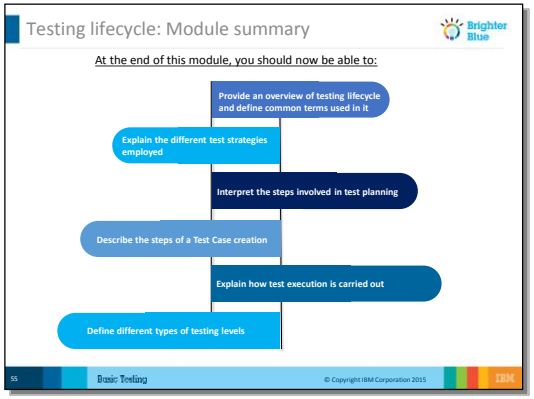
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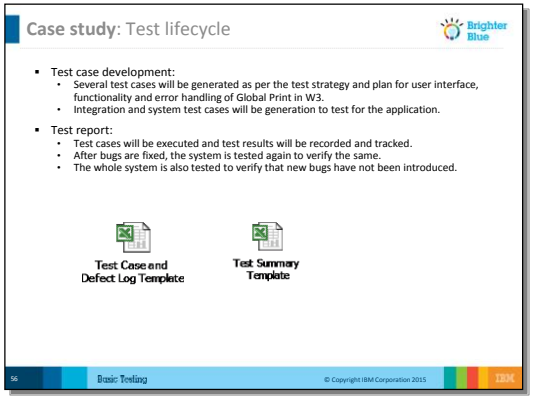
Slide Content	Use this space for your notes
Slide 50  **AIT, UAT, Structural in one** **Conversion, cutover, security in second ----- phase wise.** **It is execution of technical application, system, and database tests, such as, backup and recovery testing. Documentation tool: Excel**	

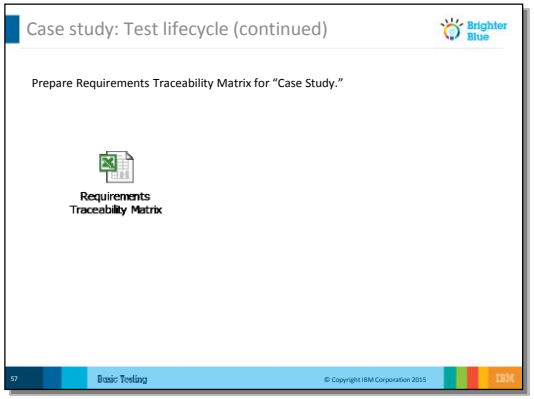
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Slide 51  **AIT, UAT, Structural in one** **Conversion, cutover, security in second ----- phase wise.**	

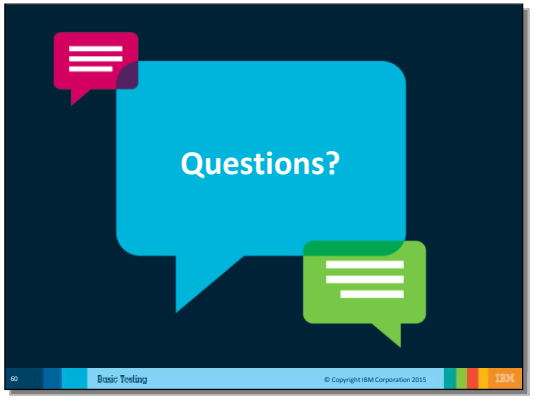
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Slide 52  **On top is a workflow of a typical defect. The highlighted boxes outline the normal defect workflow, which equals defect status in Quality Centre. The yellow boxes reflect the exception defect workflow. The dashed boxes identify the state of the defect (open or closed).**	

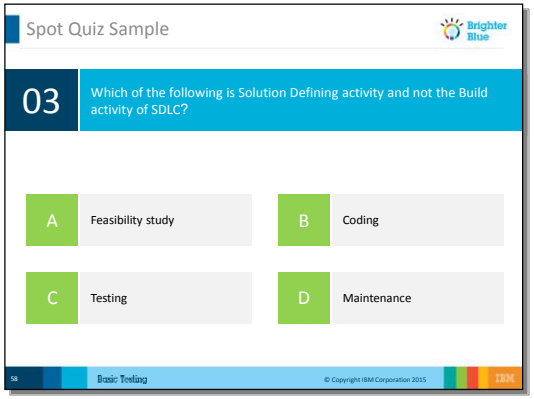
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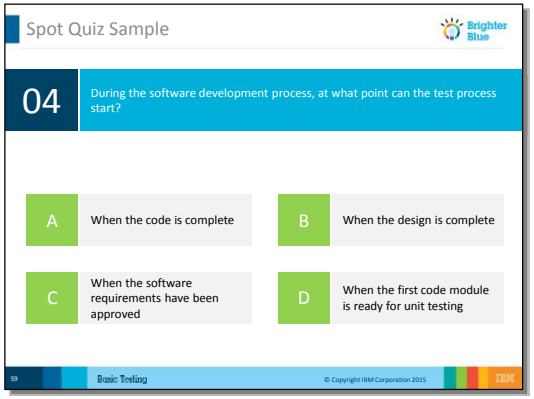
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Slide Content	Use this space for your notes
Slide 55  **Refer to the Supporting Documents_Day 1 folder for the embedded files.**	

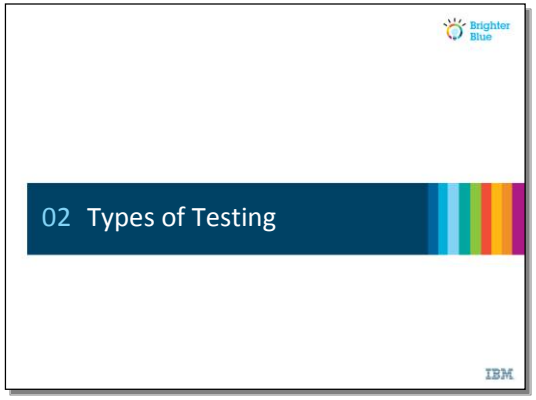
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Slide 56  **Refer to the Supporting Documents_Day 1 folder for the embedded files.**	

Slide Content	Use this space for your notes
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Slide Content	Use this space for your notes
Slide 58  **The correct answer is A. Feasibility study**	

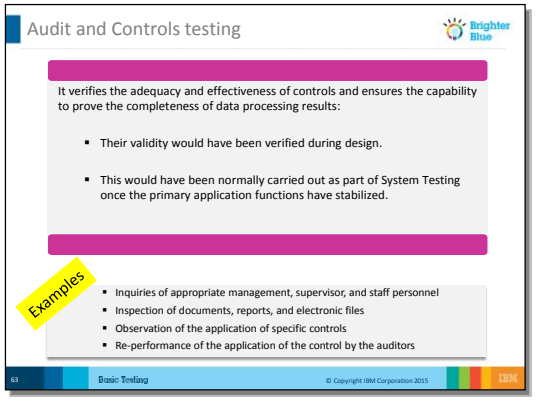
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Slide 59  **The correct answer is C. When the software requirements have been approved**	

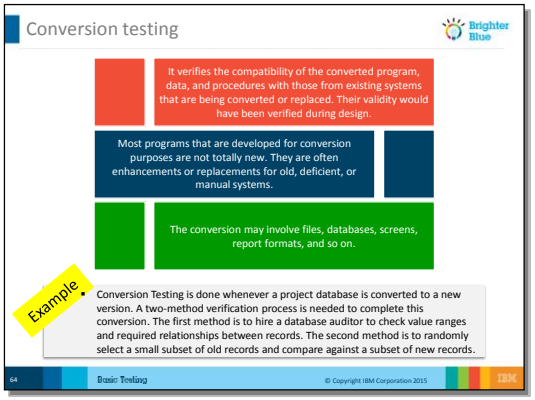
Module 2: Types and Levels of Testing

Slide Content	Use this space for your notes
Slide 60  **At the end of this module, you should be able to:** ▪ **Describe the different types of Functional testing** ▪ **Outline the different types of Structural testing** ▪ **Use examples of each type for better explanation and clarity of understanding**	

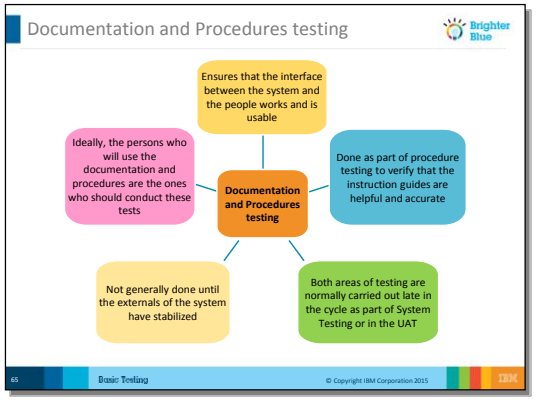
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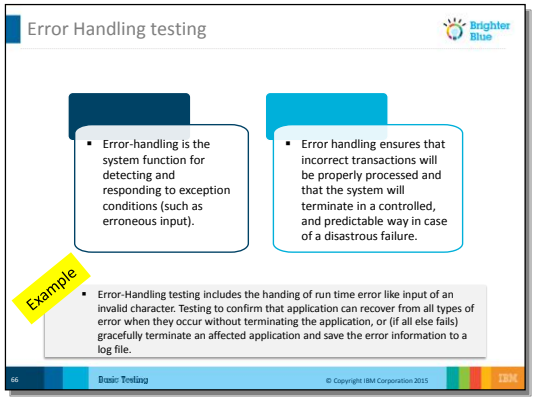
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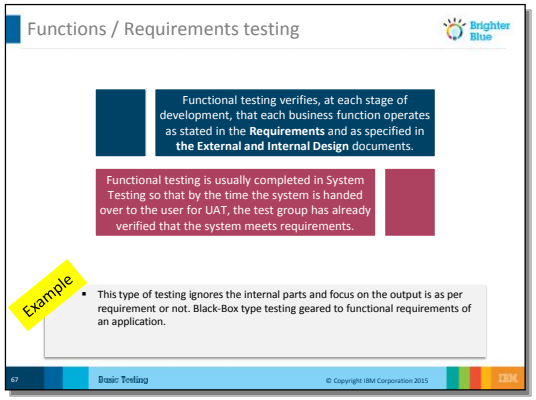
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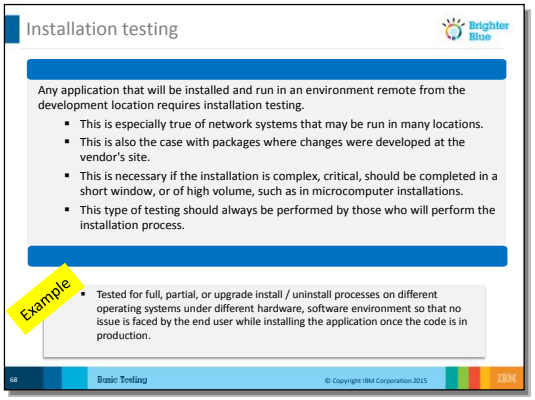
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Slide 63  **Example: Conversion Testing is done whenever a project database is converted to a new version. The First verification method involves the use of a database auditor that must be built by the development group. When ran against the converted database, the database auditor will check value ranges within a record and the required relationships between records. The second verification method involves the random selection of a small subset of old records and then a direct comparison against a corresponding subset of the new records.**	

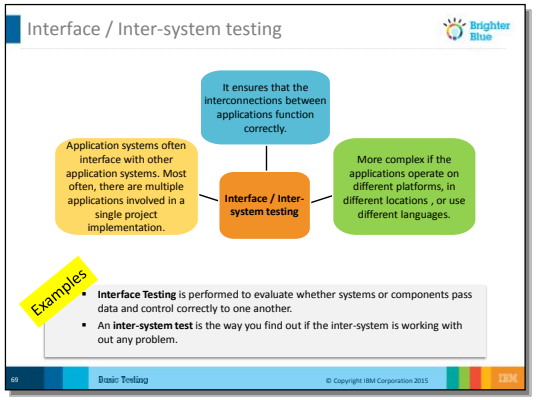
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Slide 64  **Example: Documentation is as important to a product's success as the product itself. If the documentation is poor, non-existent, or wrong, it reflects on the quality of the product. Document testing includes any type of document verification—test case specification, test incident report, test log, test plan, test procedure, and test report.**	

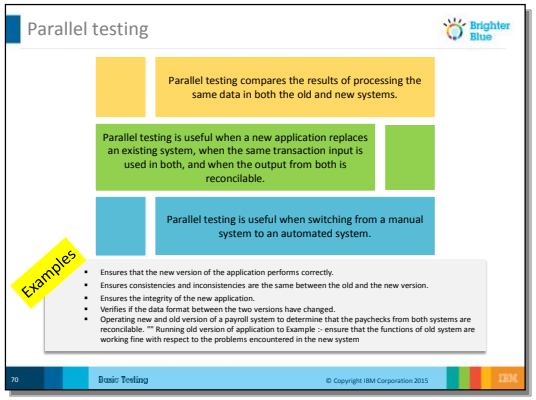
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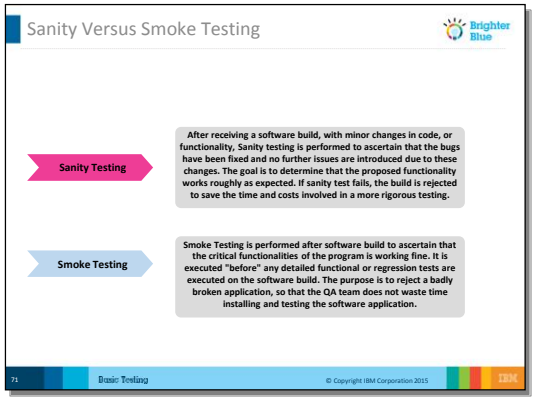
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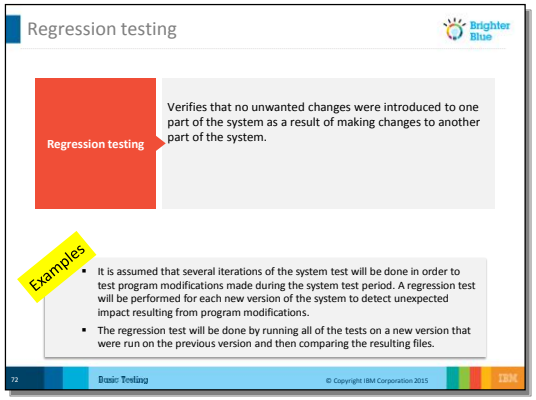
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Slide 68  **Example: Interface Testing is performed to evaluate whether systems or components pass data and control correctly to one another. It is to verify if all the interactions between these modules are working properly and errors are handled properly.** **An inter-system test, for example, could be a memory map, showing what is in memory at that moment. It basically is the way you find out if the inter-system is working without any problem, and is a big help when you do have a problem in finding a fix for it.**	

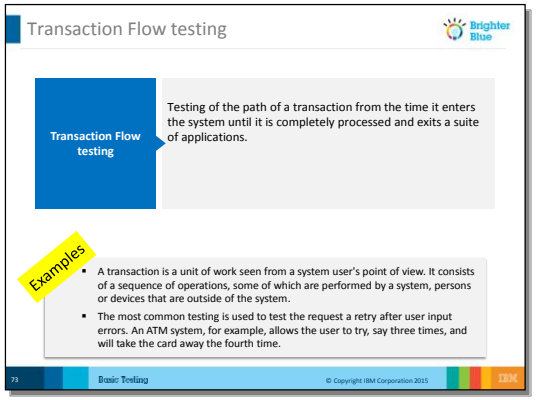
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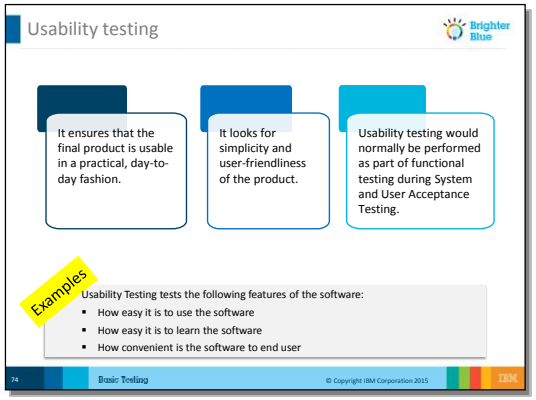
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Slide 69  **Example: Parallel testing is a testing technique in which the same inputs are entered in two different versions of the application and reporting the anomalies.** ▪ **Ensures that the new version of the application performs correctly.** ▪ **Ensures consistencies and inconsistencies are the same between the old and the new version.** ▪ **Ensures the integrity of the new application.** ▪ **Verifies if the data format between the two versions have changed.** ▪ **Operating new and old version of a payroll system to determine that the paychecks from both systems are reconcilable. "Running old version of**	

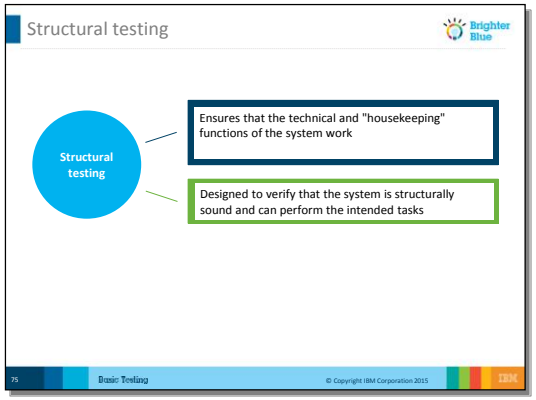
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application to Example: - ensure that the functions of old system are working fine with respect to the problems encountered in the new system.	

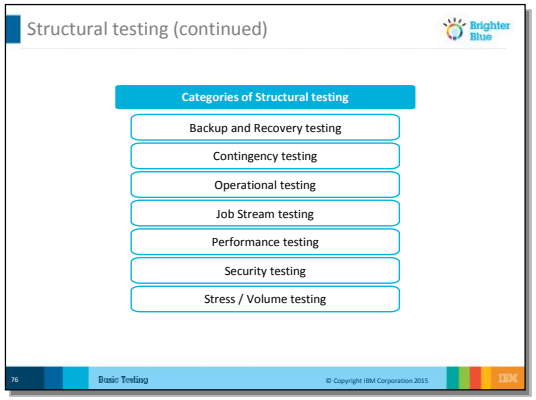
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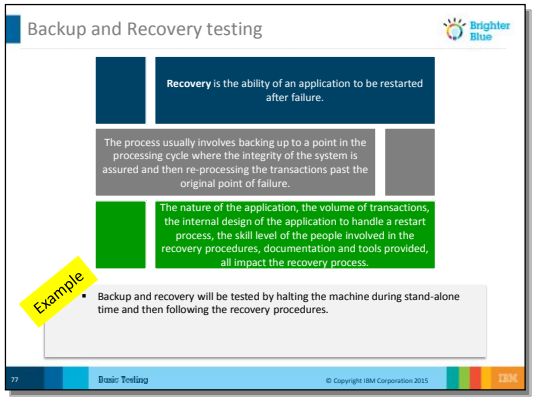
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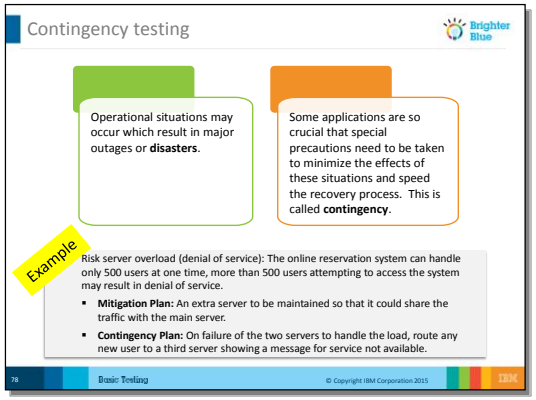
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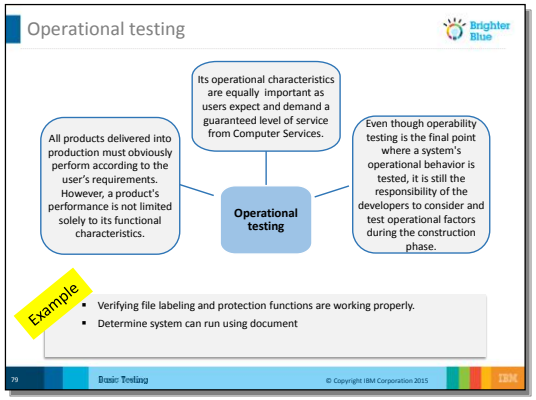
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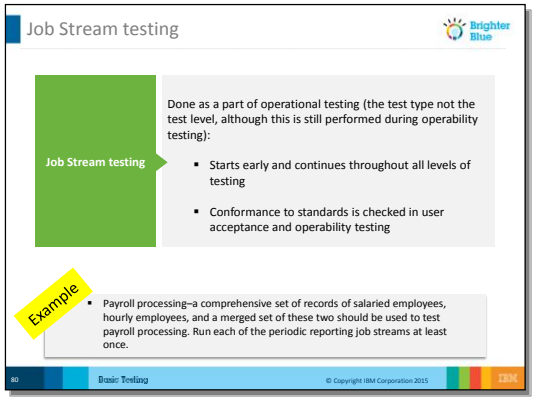
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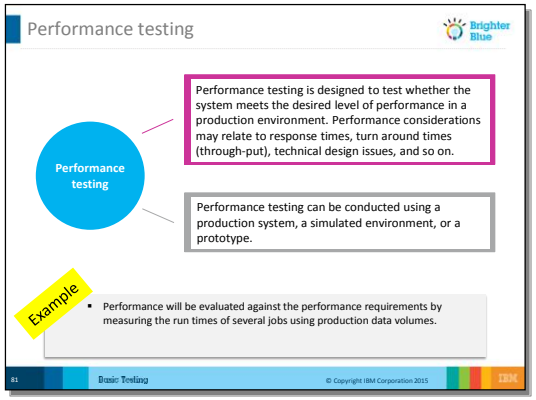
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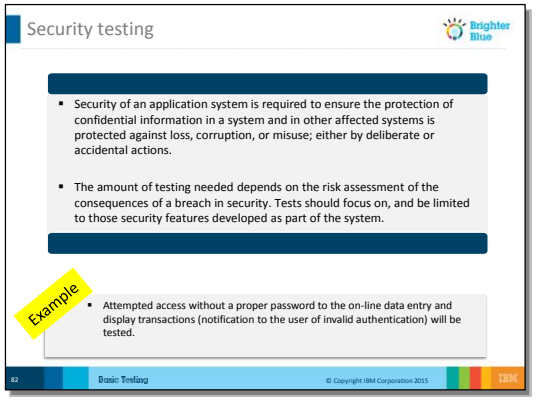
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Slide 77  **Example: The process of testing that the disaster recovery plans (also known as system contingency plans) and the associated business continuity plans work as specified.** **Risk server overload (denial of service): The online reservation system can handle only 500 users at one time, more than 500 users attempting to access the system may result in denial of service.** **Mitigation Plan: An extra server to be maintained so that it could share the traffic with the main server.**	

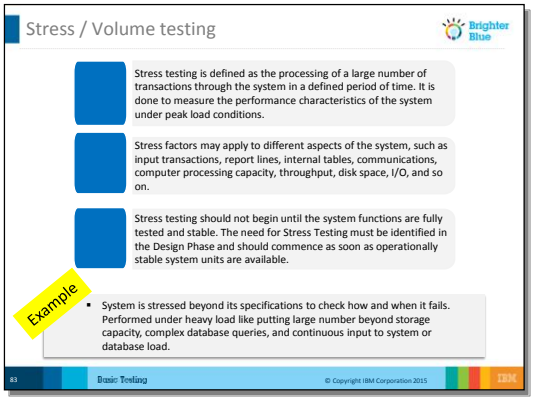
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Contingency Plan: On failure of the two servers to handle the load, route any new user to a third server showing a message for service not available.	

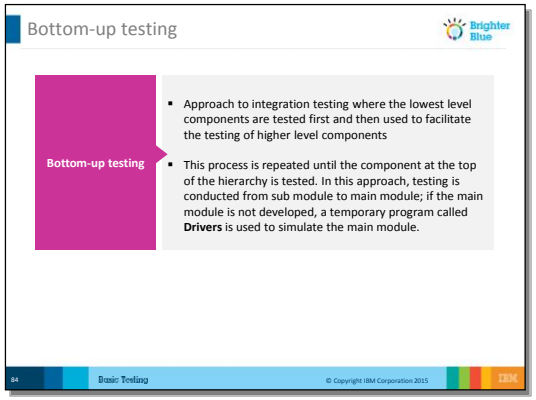
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Slide 78  **Example: To verify that before application goes to production the operating procedures and staff can properly execute the application. Evaluation of process and execution of process.** ▪ **Verifying file labeling and protection functions are working properly.** ▪ **Determine system can run using document**	

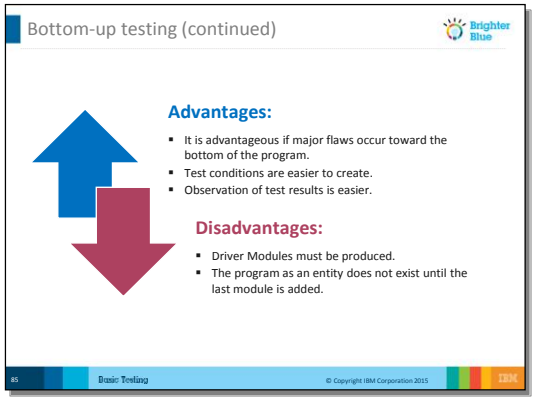
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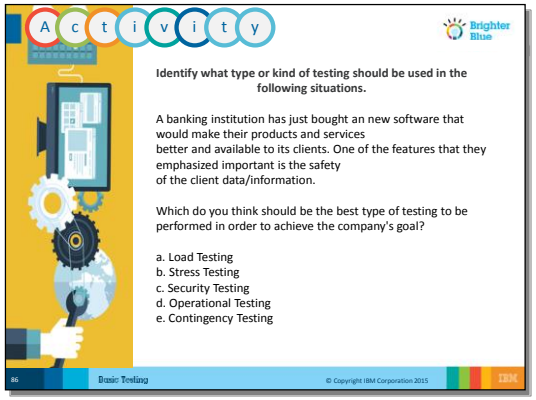
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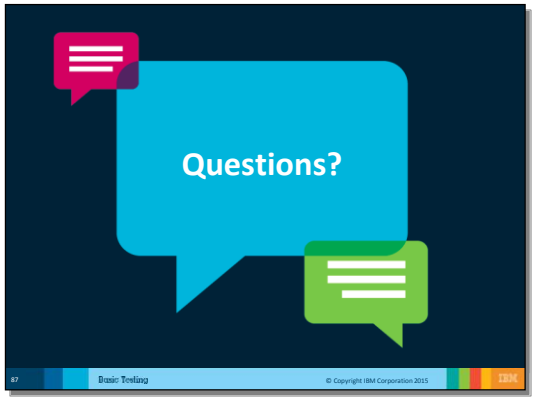
Slide Content	Use this space for your notes
Slide 81  **Example: Attempted access without a proper password to the on-line data entry and display transactions (notification to the user of invalid authentication) will be tested.** **Can system be penetrated by any hacking way? Test how well the system protects against unauthorized internal or external access. Check if system database is safe from external attacks.**	

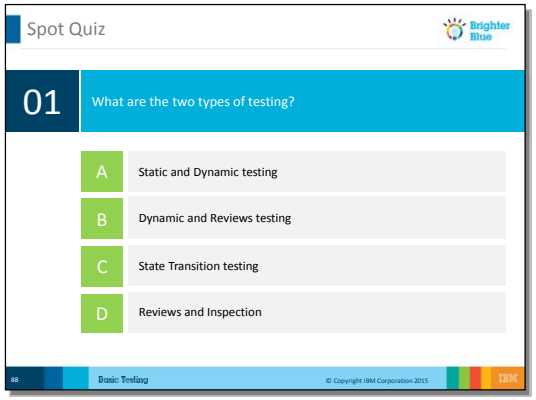
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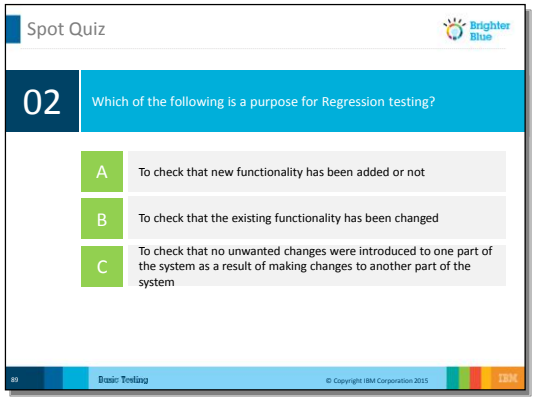
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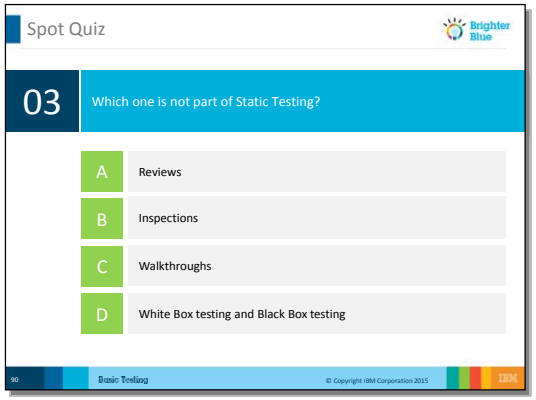
Slide Content	Use this space for your notes
Slide 84  **Example: For Unit Testing of "Sales Order Printing" program, a Driver program will have the code which will create Sales Order records using hardcoded data and then call "Sales Order Printing" program.**	

Slide Content	Use this space for your notes
Slide 85  **Identify what type or kind of testing should be used in the following situations provided by the instructor:** ▪ **Is the example given / provided require Functional or Structural testing?** **The correct answer is C. Security Testing**	

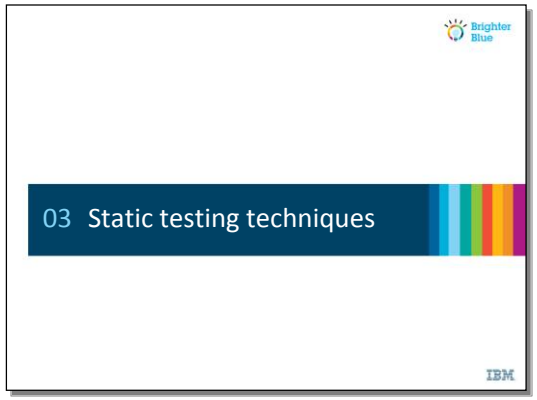
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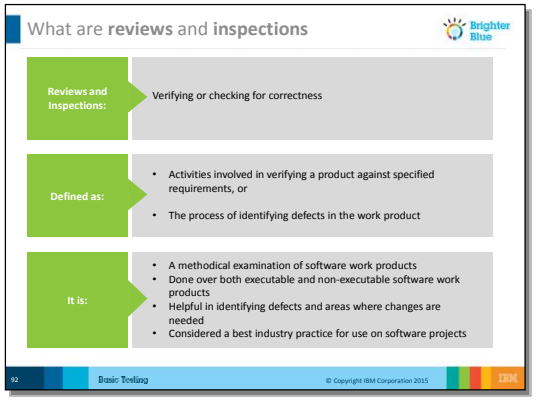
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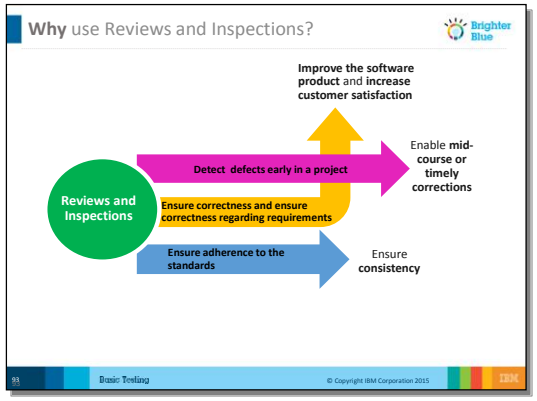
Slide Content	Use this space for your notes
Slide 89 	

Module 3: Types and Levels of Testing

Slide Content	Use this space for your notes
Slide 90  **At the end of this module, you should be able to:** ▪ **Recognize the features of reviews and inspections** ▪ **Recognize the benefits of reviews and inspections** ▪ **Recognize the steps in the reviews and inspection processes** ▪ **Show the importance of reviews and inspection through some real-life incidents** ▪ **Compare peer and facilitation reviews**	

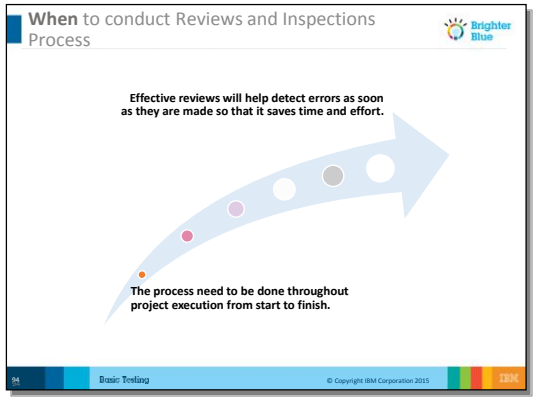
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▪ Recognize the key features of defect classification ▪ Explain how Return on Investment (ROI) helps to justify the cost of reviews	

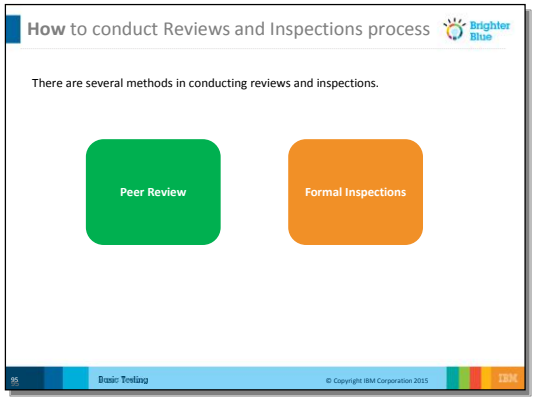
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Slide 91  **A methodical examination of software work products: Reviews and Inspections is a methodical examination of software work products generated at different phases of software development life cycle or modified during maintenance (for example, code change for a bug in an existing software).**	

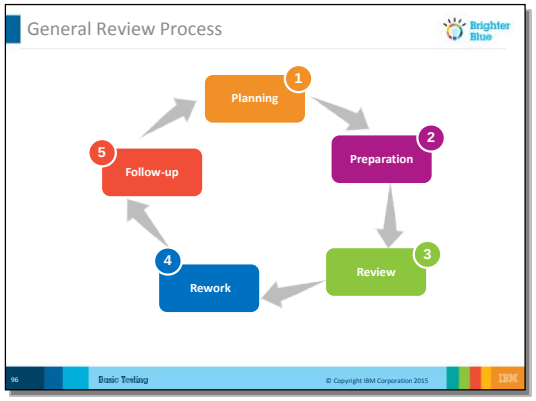
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Slide 92  **Why?** **It helps to:** • **Detect defects early in a project so as to enable mid-course or timely corrections.** • **Ensure correctness and consistency with respect to requirements, thus improving the software product and increasing customer satisfaction.** • **Ensure adherence to the standards (set by IBM or customer).**	

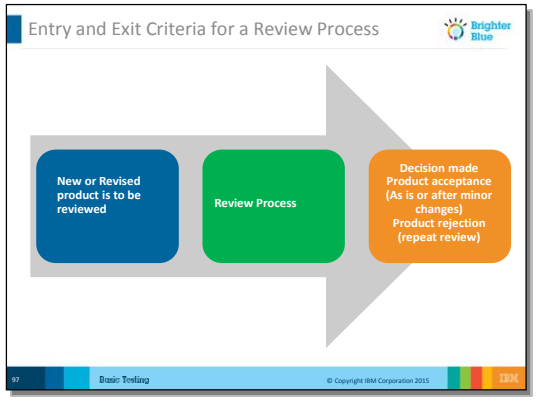
Core Testing>Basic Testing>Day 1

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Slide 94  **There are several methods involved in conducting reviews and inspections; for example, peer review and formal inspections.**	

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Slide 95  **The following illustrates the general review process.**	

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Slide 96  **The following diagram illustrates the entry and exit criteria for a review process.**	

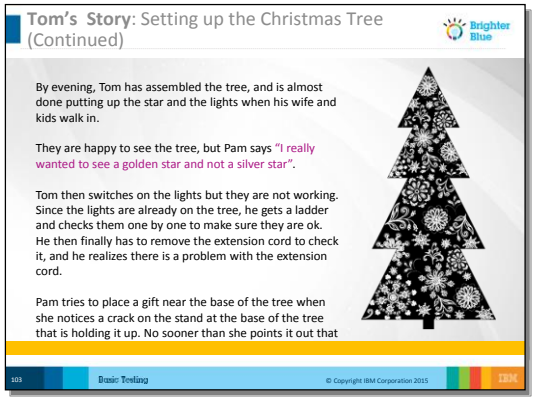
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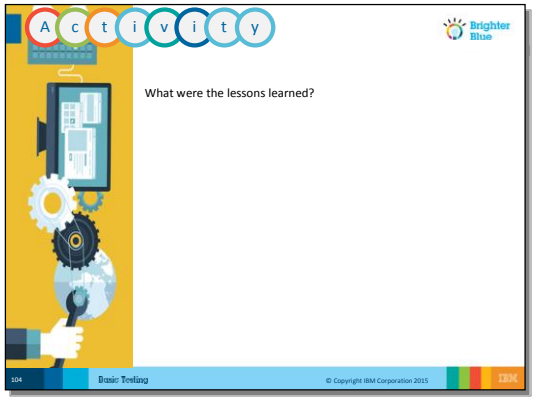
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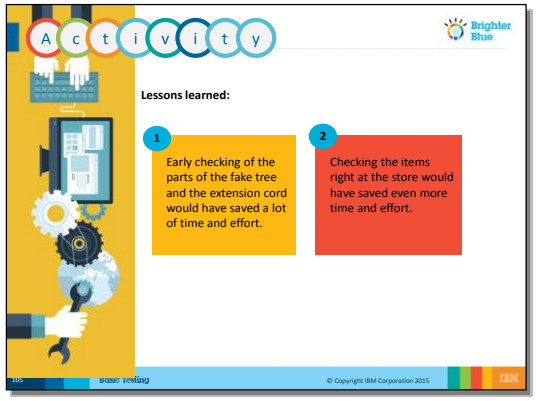
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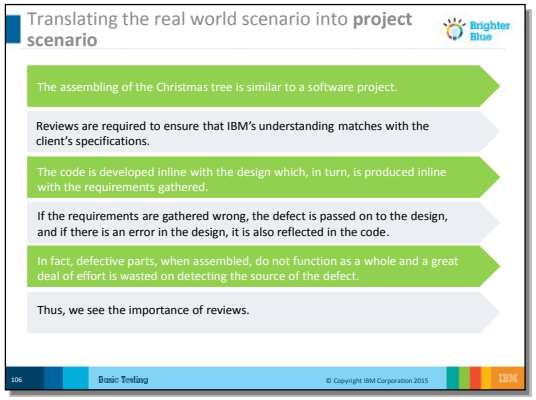
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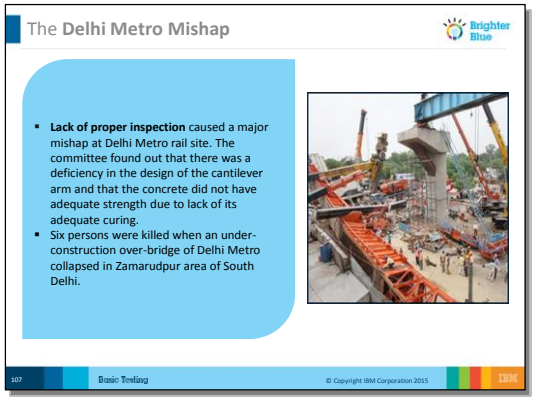
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Slide 103  **So what were the lessons learned?** **Lessons:** ▪ **Had Tom checked the parts of the fake tree before assembling it and also the extension cord before connecting it, it would have saved him a lot of time and effort. Even then, he would still have to travel all the way back to the store to get it replaced (wasted time).** ▪ **Had Tom checked the parts of the fake tree, the lights, and the extension cord right at the store, he would have saved even more time and effort.**	

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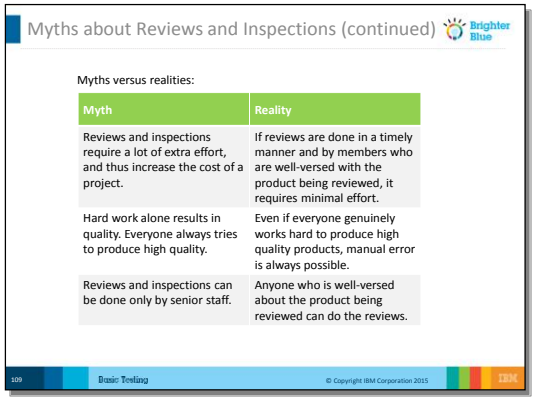
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Slide 105  ▪ **The assembling of the Christmas tree is similar to a software project. Pam giving directions to Tom is similar to a client specifying his requirements to IBM. Pam is unhappy that Tom bought a silver star instead of a gold one.** ▪ **Similarly, when clients specify requirements, reviews are required to ensure that IBM's understanding matches with the client's specifications.** ▪ **The different tree parts, lights, and extension cord are similar to different blocks of code that developers might produce. The code is developed in line with the design which in turn is produced in line with the requirements gathered.**	

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▪ If the requirements are gathered wrong, the defect is passed on to the design, and if there is an error in the design, it is also reflected in the code. So when the code is tested, desired results are not achieved. ▪ In fact, defective parts, when assembled, do not function as a whole, and a great deal of effort is wasted on detecting the source of the defect, that is, is it a defect in the code or a defect in the design or a defect in the requirements gathered? ▪ Thus, we see the importance of reviews and also the importance of detecting errors as soon as they are made.	

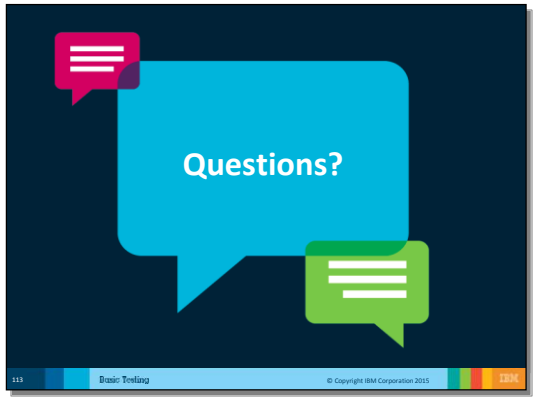
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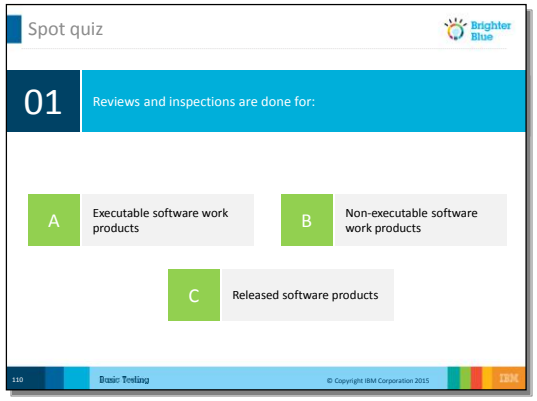
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Slide 107  ▪ **The main reason of such accidents is lack of supervision, which leads to poor quality construction.** ▪ **There are certain operating procedures which have to be followed at every step of construction and inspection. As these types of projects depend upon team work, it is the moral duty of everyone, either director or laborer, to not compromise on the quality and bypass inspection process.** ▪ **Since Gammon and DMRC had bypassed certain checks and inspection processes, this disaster occurred.**	

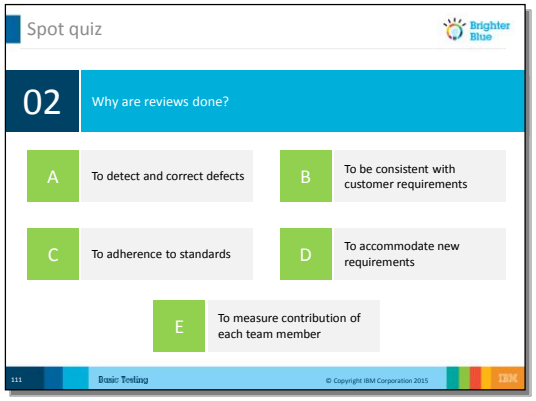
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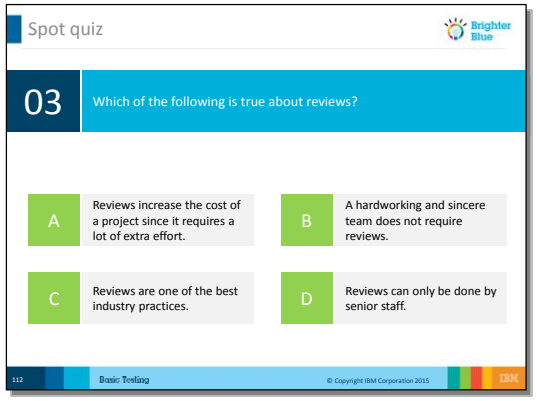
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Slide 108  **Reality:** ▪ **If reviews are done in a timely manner and by members who are well-versed with the product being reviewed, it requires minimal effort. No reviews result in defects and rework taking even more time and effort to make fixes.** ▪ **Even if everyone genuinely works hard to produce high quality products, manual error is always possible. Also, it is easy to miss defects when we continuously look at the same piece of code. A peer might be able to catch it easily.** ▪ **Anyone who is well-versed about the product being reviewed can do the reviews.**	

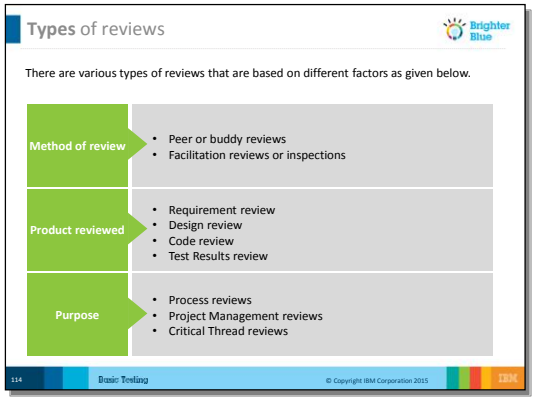
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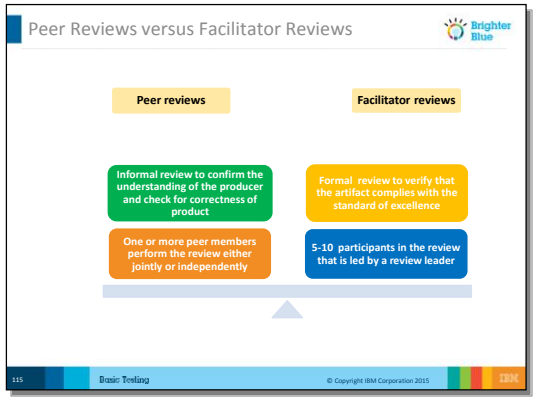
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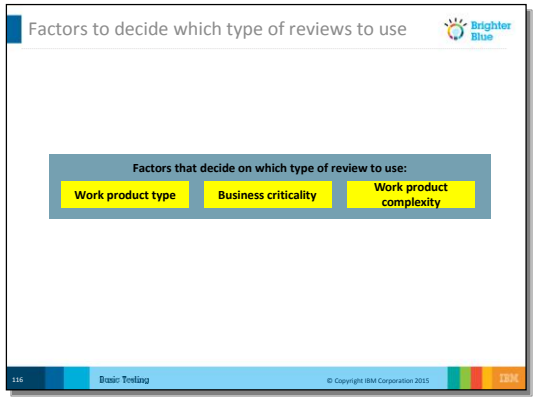
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Slide 113  **There are many types of reviews based on different factors. A few are mentioned here:** **Based on method of review:** • **Peer or buddy reviews** • **Facilitation reviews or inspections** **Based on product reviewed:** • **Requirement review** • **Design review**	

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• Code review • Test Results review Based on purpose: • Process reviews (to check compliance to organization processes) • Project Management Reviews (to review project health and check risks) • Critical Thread Reviews (meetings to discuss and review topics on people, process, and performance)	

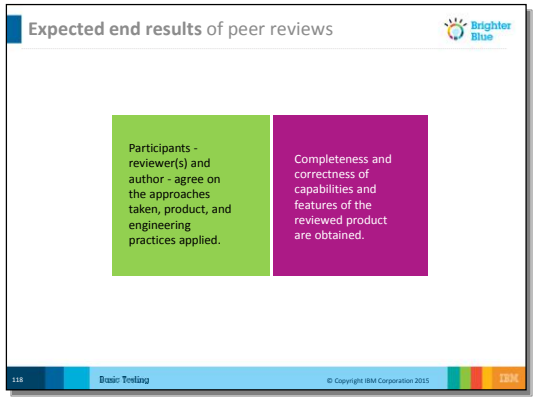
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Slide 114  ▪ **Peer reviews** ○ **Peer or buddy reviews (walkthroughs) are informal reviews used to confirm the understanding of the producer and check for correctness of product being reviewed.** ○ **One or more peer members (with expertise in the subject of the product to be reviewed) perform the review either jointly or independently.** ▪ **Facilitator Reviews**	

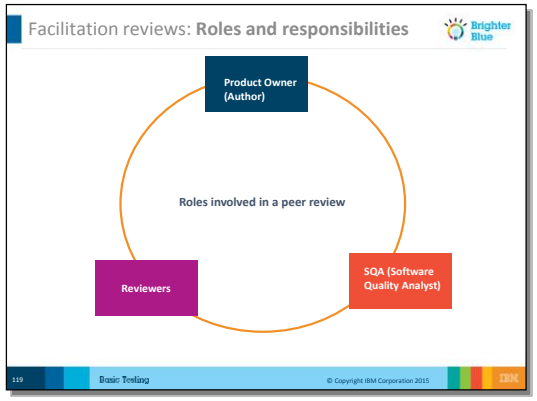
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○ **Facilitator reviews (inspection) are formal reviews used to verify that the artifact compiles with the standard of excellence.** ○ **A team of not less than 5 percent and not more than 10 (each with specific defined roles) participates in the review that is led by a review leader.**	

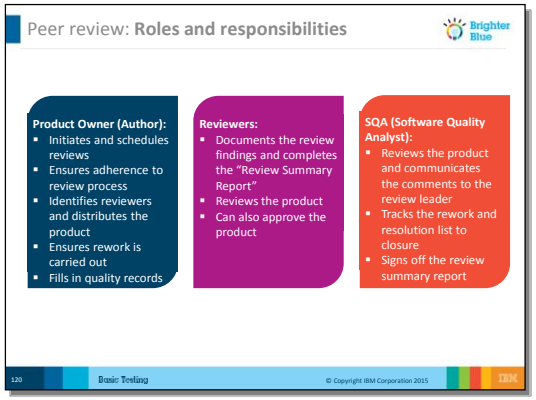
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Slide 115  **Work product type - Design document, code, or any other**	

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Slide 116  **Peer reviews:** ▪ **This is a less formal verification technique - This is a less formal verification technique in which a technical peer analyzes or improves the quality of the original work product, for example, by finding defects or proposing a more optimal solution.** ▪ **It is the author who initiates the session - Since peer reviews cater to the needs of the producer or author of the software artifact (in acquiring superior knowledge of all aspects of it), it is the author who initiates the session.**	

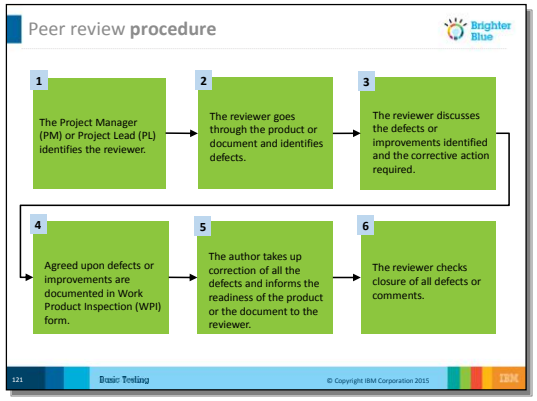
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▪ There may be several peer reviews in each software development life cycle activity - There may be several peer reviews in each software development life cycle activity and they yield open issues and action items. While these issues and action items may be tracked to closure, the only measurement taken from these reviews is a count of the number of peer reviews held along with effort spent and process improvements identified.	

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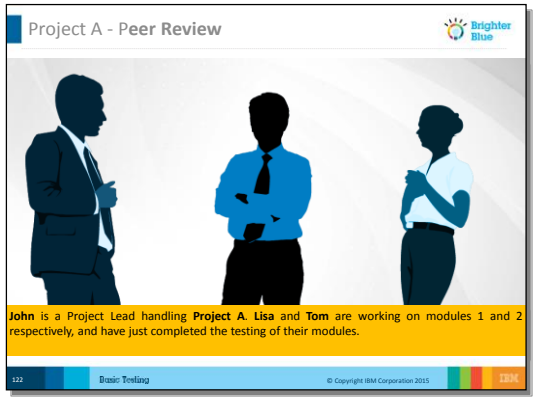
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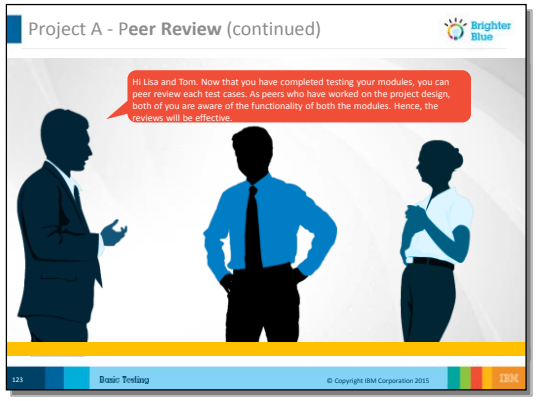
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Slide 119  **Product Owner (Author):** ▪ **Initiating the review process and scheduling reviews** ▪ **Ensuring the review process is adhered to** ▪ **Identifying reviewers and distributing the product for review. The owner may seek help from his manager in identification.** ▪ **Ensuring any rework as a result of the review is carried out** ▪ **Filing the quality records that are generated during the review process as objective evidence of having carried out the review** **Reviewers:**	

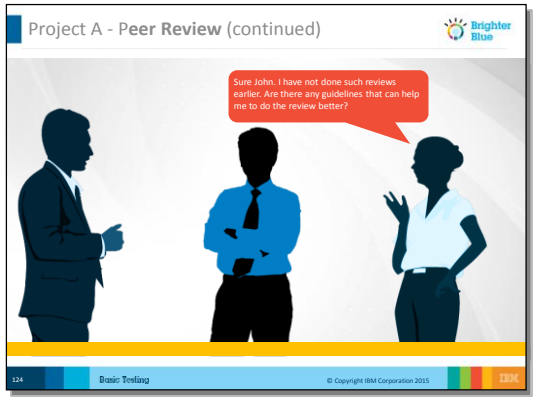
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▪ Documenting the review findings and completing the 'Review Summary Report' ▪ Reviewing the product ▪ Can also approve the product SQA (Software Quality Analyst): ▪ Where applicable, reviewing the product and communicating the comments to the review leader ▪ Where applicable, tracking the rework and resolution list to closure ▪ Signing off the review summary report after ensuring the defect / resolutions list is closed and the summary report is complete	

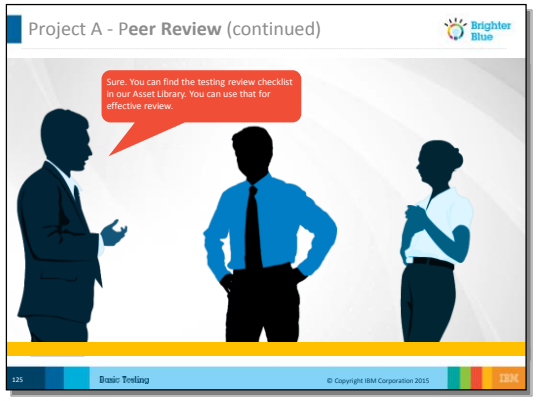
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Slide 120  **•The Project Manager (PM) or Project Lead (PL) identifies the reviewer based on the size, complexity, and importance of the product or document to be reviewed.** **•The reviewer goes through the product or document and identifies defects based on appropriate checklists or specifications or standards.** **•The reviewer discusses the defects or improvements identified, the corrective action required, and provides clarifications if any.** **•Once the defects or improvements are agreed upon, they are documented in the Work Product Inspection (WPI) form. It is the**	

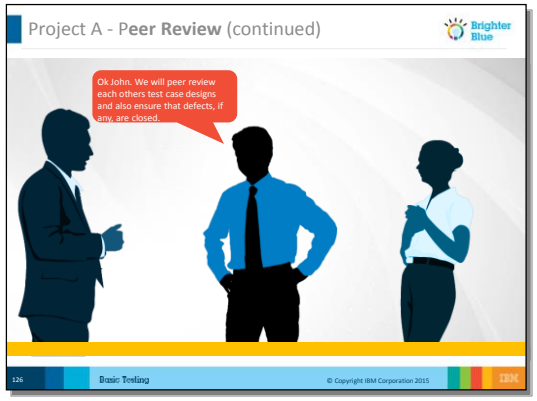
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responsibility of the PM or PL to ensure that the reviews are conducted, results documented, and the defect corrections are carried out. •The author takes up the correction of all the defects identified to be corrected and informs the readiness of the product or the document to the reviewer. •If re-review is required, another iteration of review is done, but if the review result requires only verification of the closure of defects or comments, the reviewer checks whether all defects have been closed.	

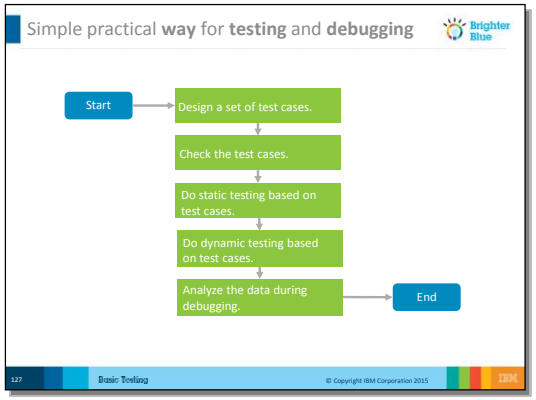
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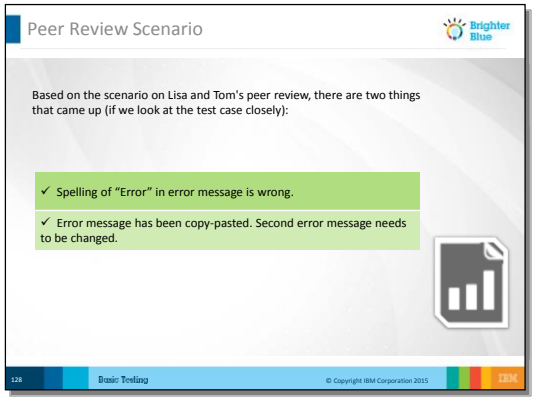
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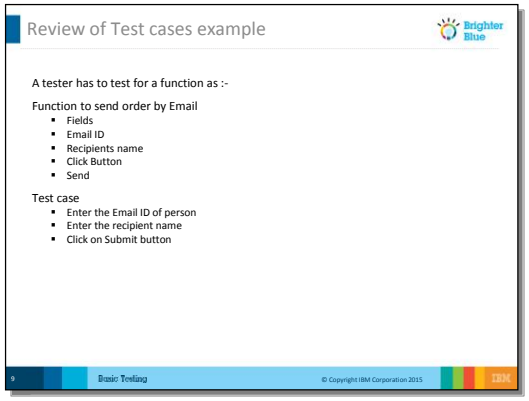
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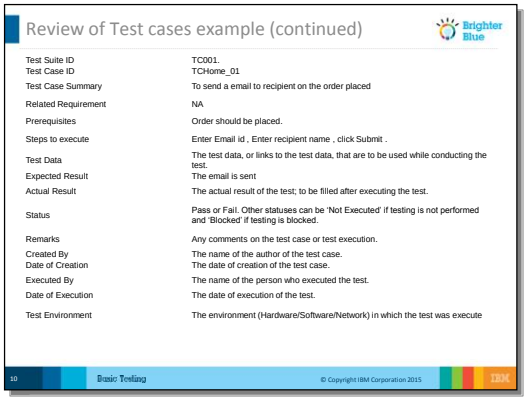
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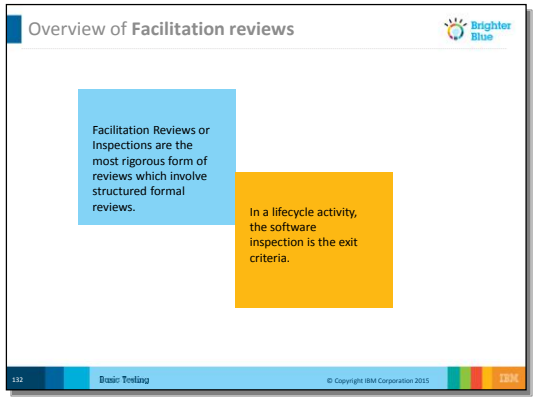
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Slide 127  **Based on the scenario on Lisa and Tom's peer review, there are two things that came up (if we look at the code closely):** • **Spelling of 'Error' in error message is wrong.** • **Error message has been copy-pasted. Second error message needs to be changed.**	

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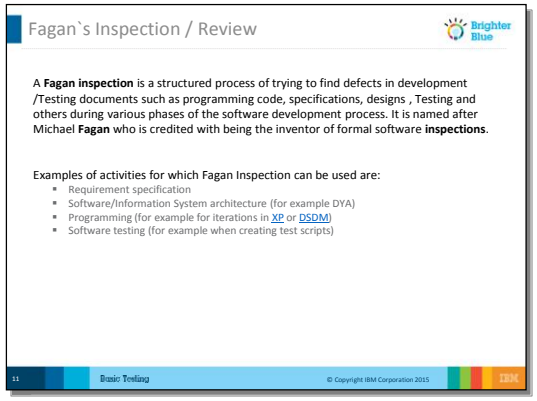
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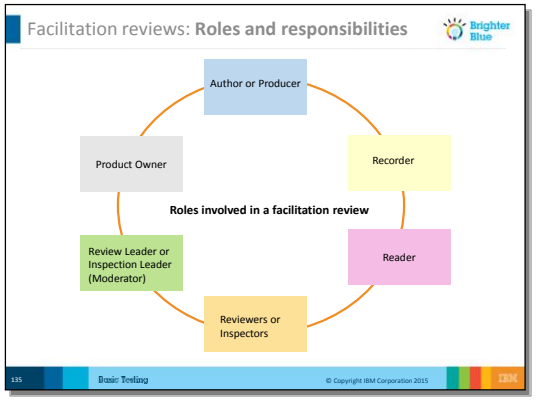
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Slide 131  ▪ **Facilitation Reviews or Inspections (also called Fagan's Review) are the most rigorous form of reviews which involve structured formal reviews done by a group of technical personnel who analyze or improve the quality of the original work product as well as the quality of the method.** ▪ **In a lifecycle activity, the software inspection is the exit criteria or gate that concludes the activity.**	

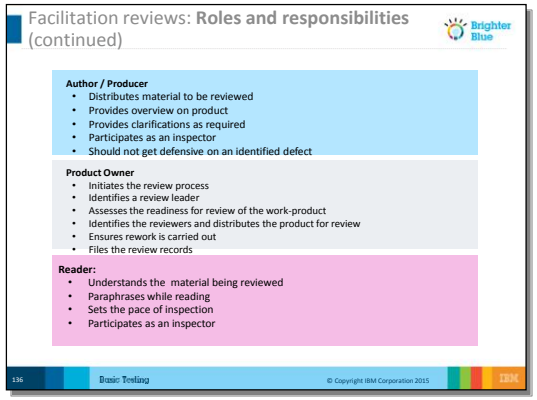
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Slide 132  **Characteristics of Facilitation reviews** • **They are done by technical people for technical people.** • **They consist of structured, well defined processes with well-defined roles for participants.** • **Reviewers are prepared in advance, and clarifications are obtained before the meeting.** • **Focus is on identifying problems and not resolving them.** • **Review data from these reviews is recorded and used for monitoring effectiveness of the review procedure.**	

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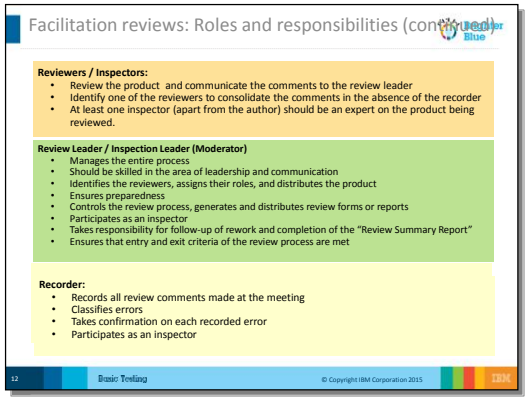
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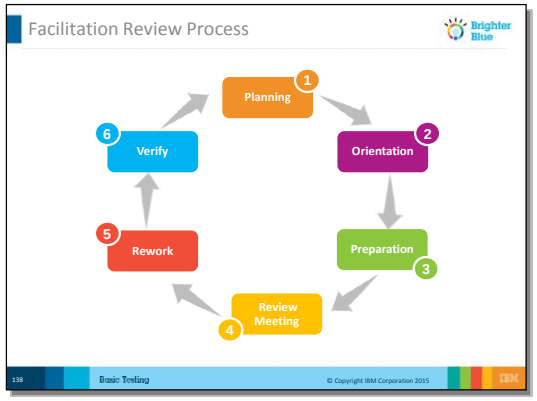
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Slide 135  **Author / Producer:** • **Distributes material to be reviewed** • **Provides overview on product** • **Provides clarifications as required** • **Participates as an inspector** • **Should not get defensive on an identified defect** **Product Owner:** • **Initiates the review process** • **Identifies a review leader; may seek help from his manager in identifying a review leader**	

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• Assesses the readiness for review of the work-product along with the review leader • Identifies the reviewers and distributes the product for review along with the review leader • Ensures that any rework as a result of the review is carried out • Files the review records that are generated during the review process as objective evidence of having carried out the review • Review Leader / Inspection Leader (Moderator): • Manages the entire process and ensures that the review process is adhered to • Should be skilled in the area of leadership and communication, and be impartial • Identifies the reviewers and assigns their roles, distributes the product and related documents • Ensures preparedness before the review • Controls the pace of review meetings, moderates the review, and also generates and distributes review forms or reports • Participates as an inspector • Takes responsibility for follow-up of rework and completion of the 'Review Summary Report' • Ensures that entry and exit criteria of the review process are met	

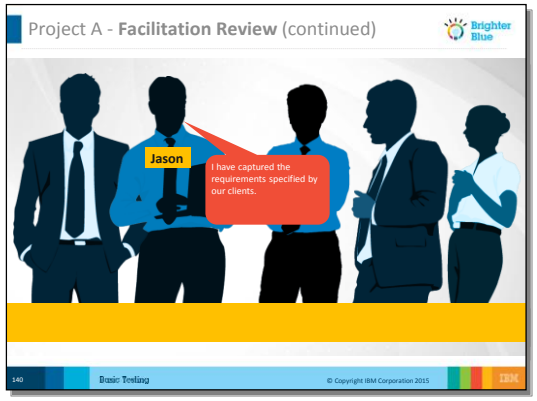
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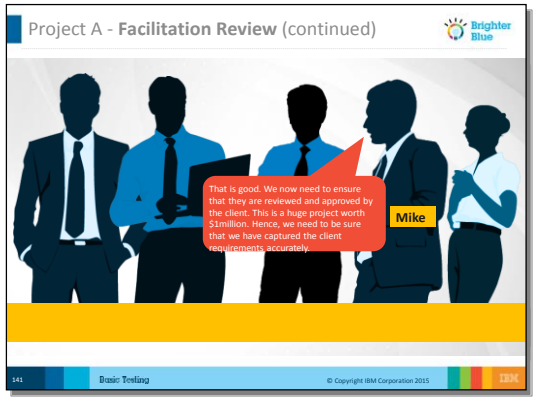
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Slide 136  **Reviewers / Inspectors:** ■ **Review the product and communicate the comments to the review leader in the review meeting. This involves:** ○ **Preparing for the inspection meeting** ○ **Understanding the product** ○ **Doing the inspection** ○ **Focusing on product and NOT the producer**	

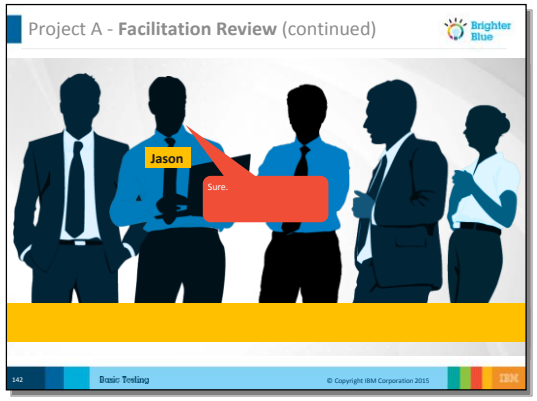
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▪ In the absence of the recorder, the review leader is responsible for identifying one of the reviewers to consolidate the comments. ▪ Everybody involved in the facilitation review (leader, reader, recorded, and author) is an inspector, but it is important that at least one inspector (apart from the author) should be an expert on the product being reviewed. **Reader:** ▪ Understands the material being reviewed ▪ Paraphrases while reading ▪ Sets the pace of inspection ▪ Participates as an inspector **Recorder:** ▪ Records all review comments made at the meeting (as indicated by leader) ▪ Classifies errors ▪ Takes confirmation on each recorded error ▪ Participates as an inspector	

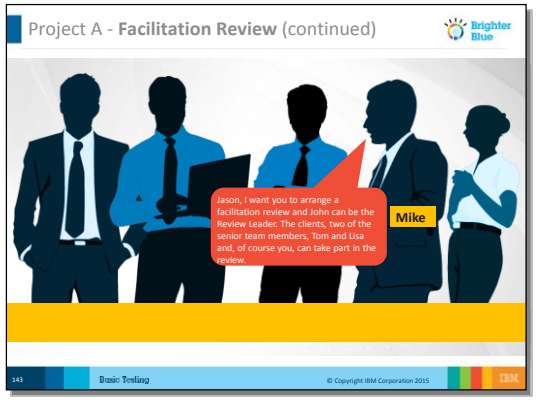
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Slide 137  ▪ Planning: Choose team, materials, and dates ▪ Orientation: Present product, process and goals (Kick-off) ▪ Preparation: Check product, and note issues ▪ Review Meeting: Consolidate issues ▪ Rework: Correct defects ▪ Verify: Verify product / process quality (Follow-up)	

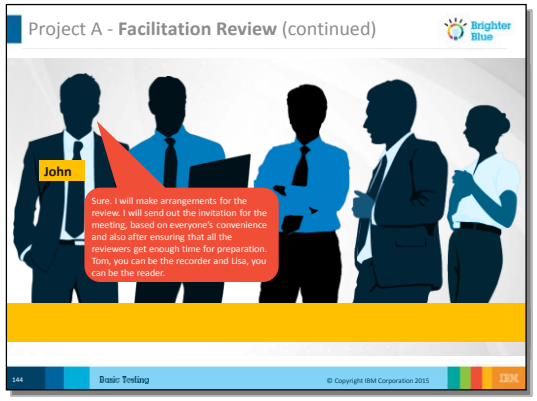
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Slide 138  **The clients have just specified their requirement to the onsite coordinator, Jason. After which, he went to the site to meet up with the team to discuss it, to ensure that everything as expected is covered and nothing gets missed out.**	

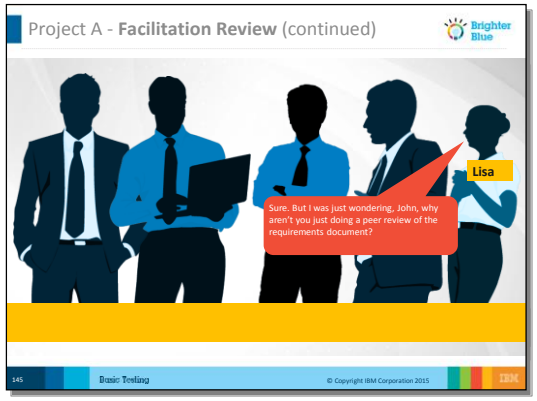
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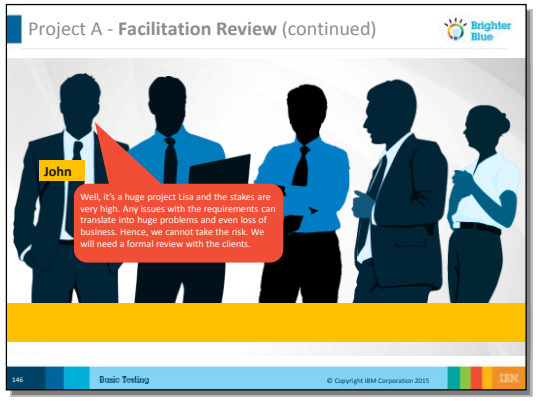
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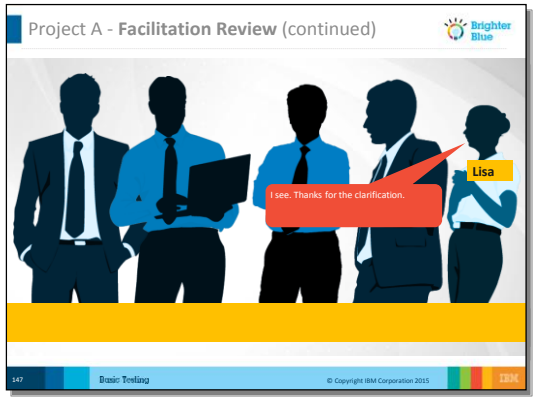
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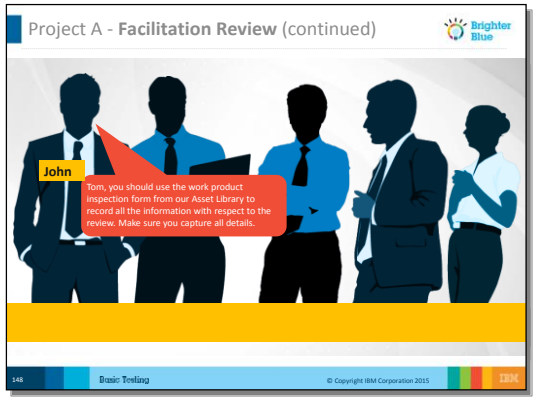
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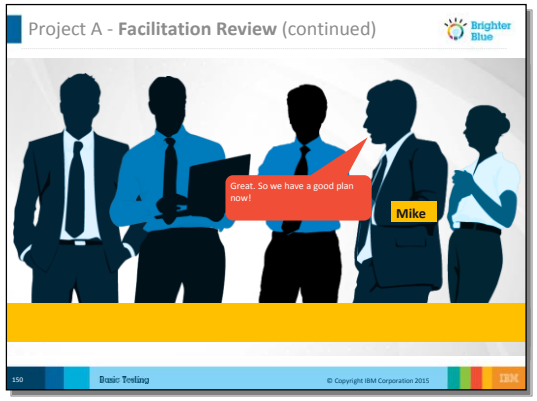
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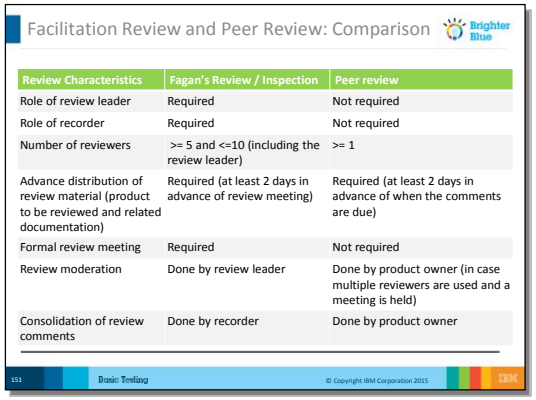
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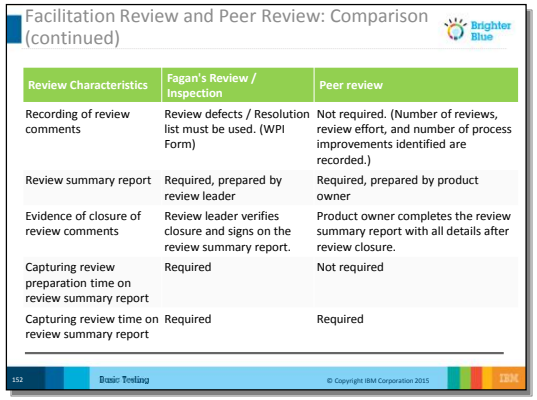
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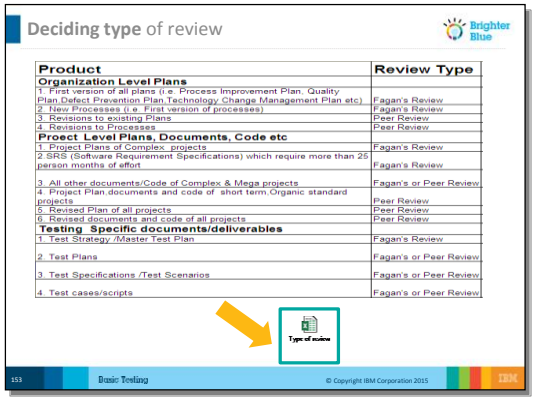
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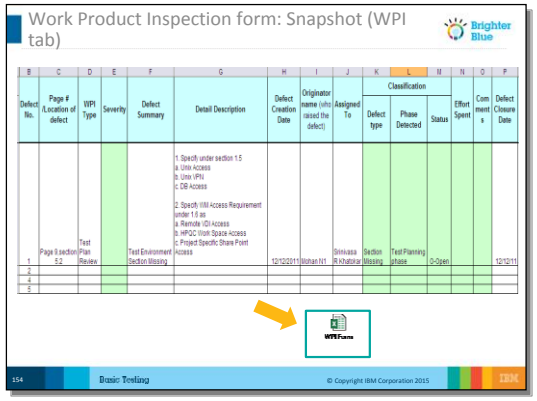
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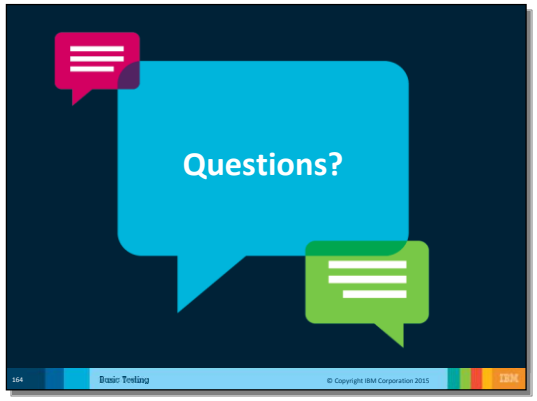
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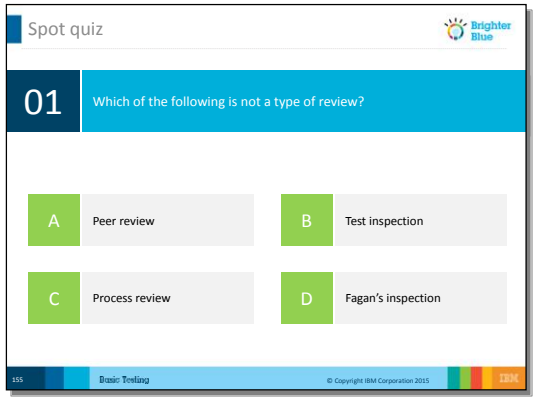
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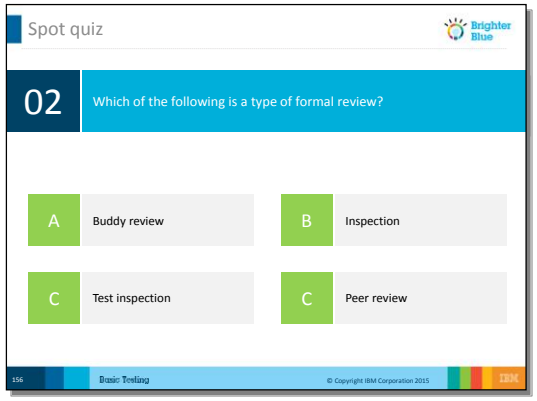
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Slide 152  **Refer to the Supporting Documents_Day 1 folder for the embedded file.**	

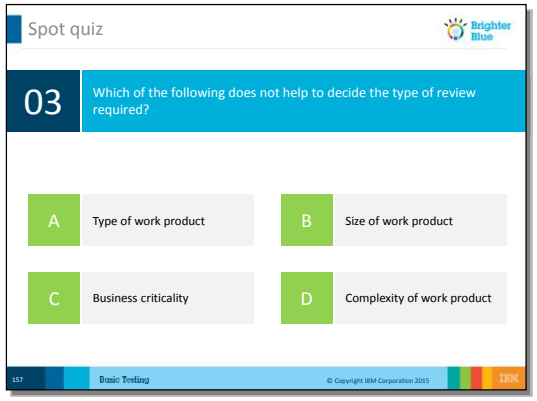
Product	Review Type
Organization Level Plans	
1. First version of all plans (i.e. Process Improvement Plan, Quality Plan,Defect Prevention Plan,Technology Change Management Plan etc)	Fagan's Review
2. New Processes (i.e. First version of processes)	Fagan's Review
3. Revisions to existing Plans	Peer Review
4. Revisions to Processes	Peer Review
Proect Level Plans, Documents, Code etc	
1. Project Plans of Complex projects	Fagan's Review
2.SRS (Software Requirement Specifications) which require more than 25 person months of effort	Fagan's Review
3. All other documents/Code of Complex & Mega projects	Fagan's or Peer Review
4. Project Plan,documents and code of short term,Organic standard projects	Peer Review
5. Revised Plan of all projects	Peer Review
6. Revised documents and code of all projects	Peer Review
Testing Specific documents/deliverables	
1. Test Strategy /Master Test Plan	Fagan's Review
2. Test Plans	Fagan's or Peer Review
3. Test Specifications /Test Scenarios	Fagan's or Peer Review
4. Test cases/scripts	Fagan's or Peer Review

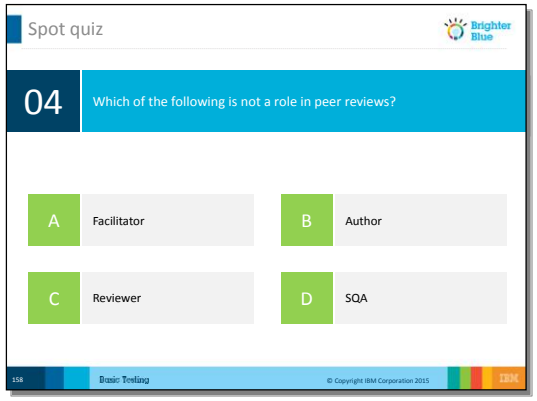
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Slide 153  **Refer to the Supporting Documents_Day 1 folder for the embedded file.**	

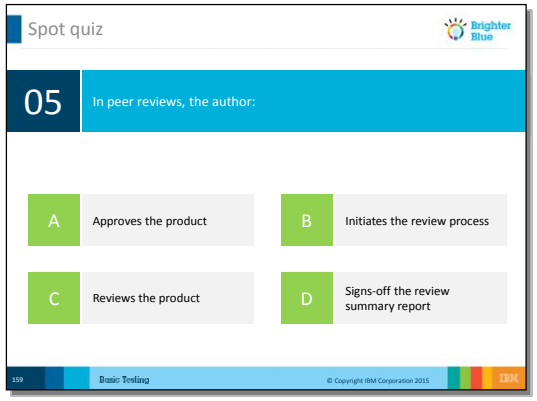
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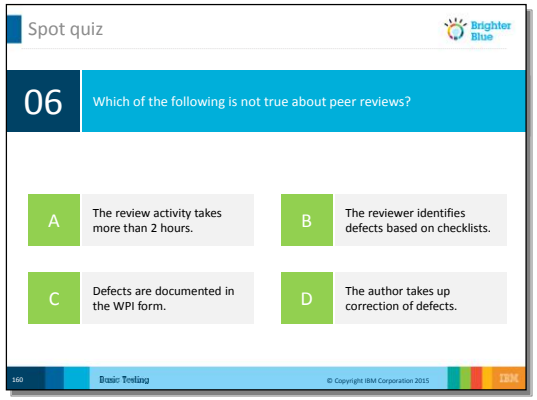
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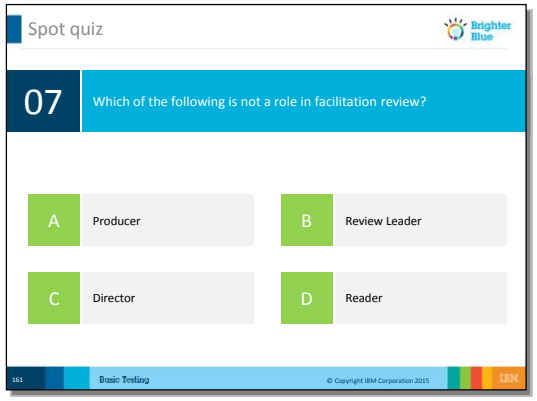
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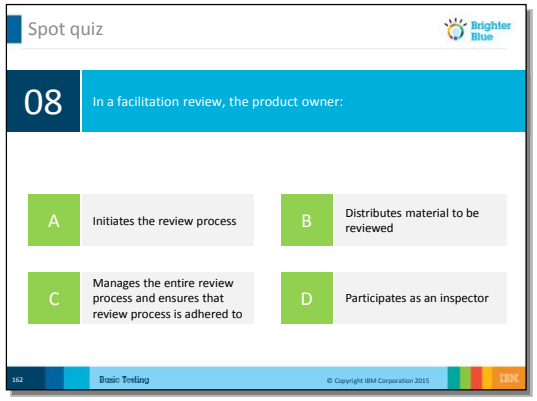
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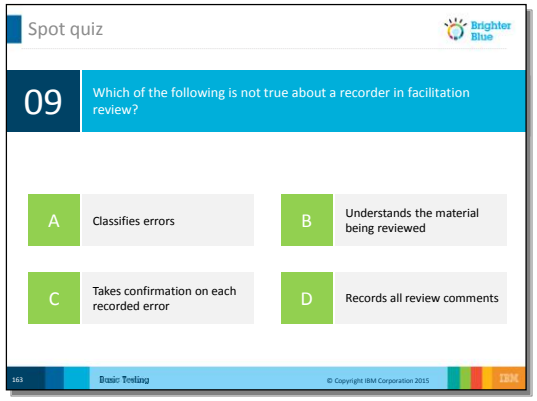
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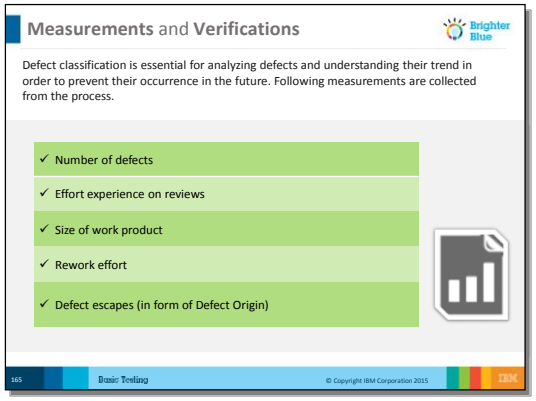
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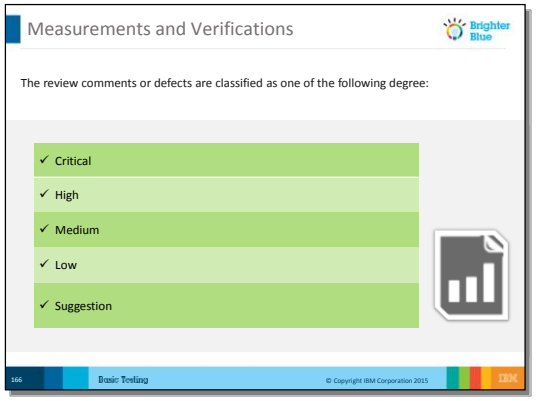
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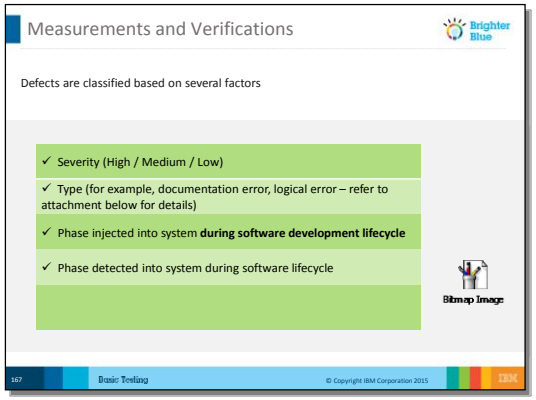
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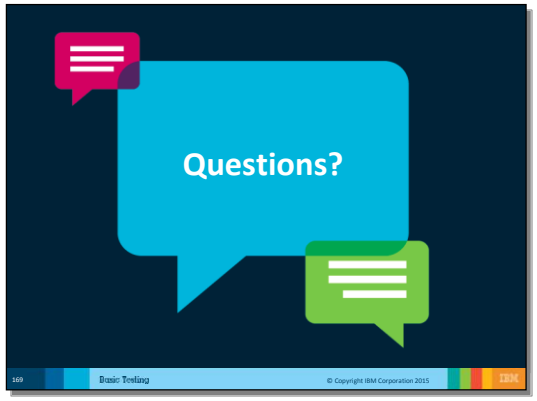
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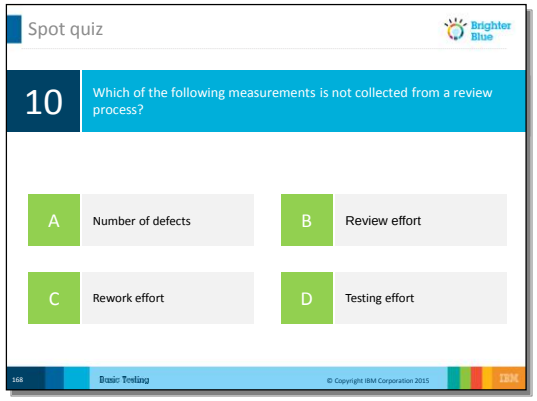
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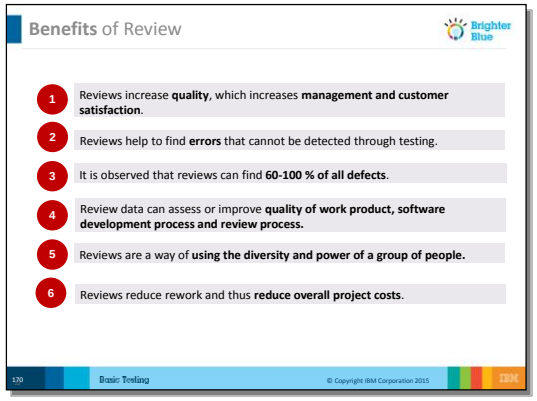
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Slide 166  **Refer to the Supporting Documents_Day 1 folder for the embedded image.**	

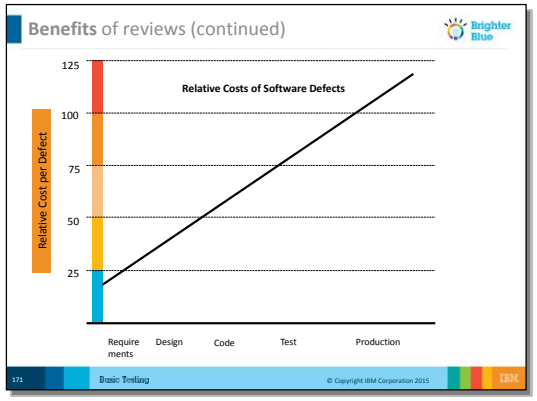
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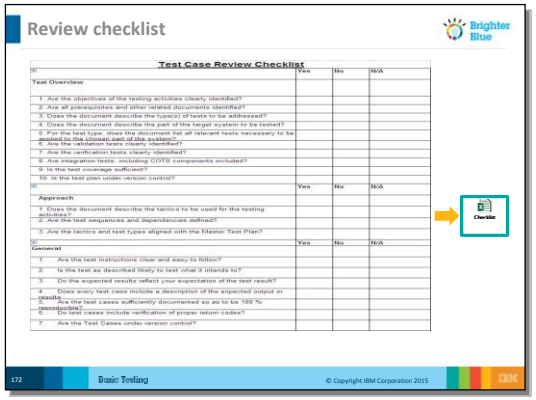
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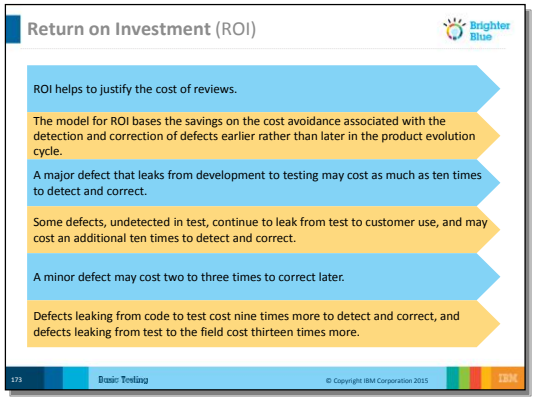
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Slide 169  ▪ **Reviews increase quality–which increases management and customer satisfaction.** ▪ **Reviews help to find errors that cannot be detected through testing.** ▪ **It is observed that reviews can find 60-100% of all defects.** ▪ **Review data can assess or improve quality of:** ○ **Work product** ○ **Software development process** ○ **Review process**	

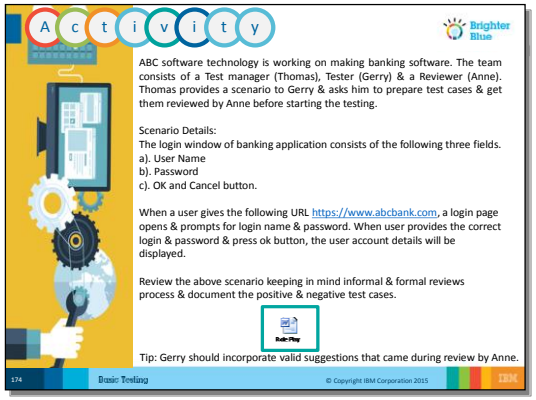
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▪ Reviews are a way of using the diversity and power of a group of people to: ○ Detect defects (errors) ○ Remove defects as close to the point of insertion as possible ○ Determine product progress or status ○ Identify potential improvements ○ Produce technical work of a more uniform and predictable quality ○ Gain ownership by the project team ○ Establish an audit trail from systems requirements allocated to software through the successive phases of development (traceability) ▪ Reviews reduce rework and thus reduce overall project cost. ○ Rework accounts for 44% of development cost while review accounts for only 15%. ○ The 44% is split as: Requirement (1%), Design (12%), Coding (12%), and Testing (19%).	

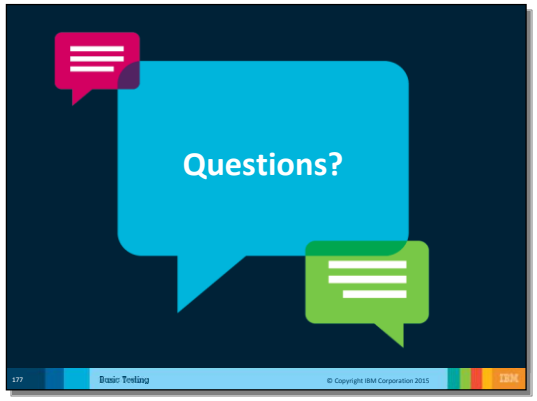
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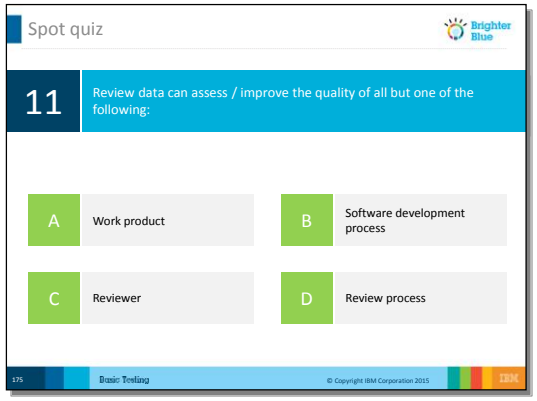
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Slide 170  **It is observed that the maximum number of defects are injected during requirements phase itself (Requirements – 56%, Design – 27%, Code – 7%, Other – 10%). Upstream defect removal is 10 to 100 times cheaper. Hence, timely reviews help in keeping project cost under control.**	

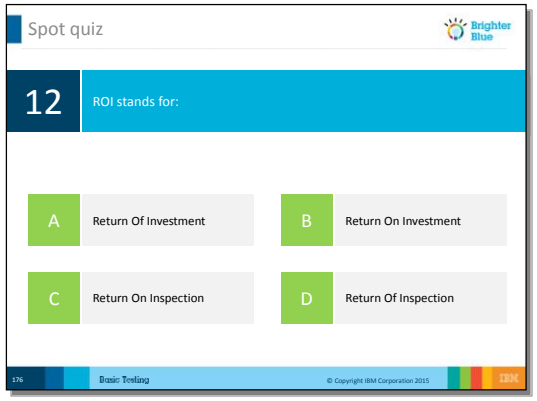
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Slide 171  **Refer to the Supporting Documents_Day 1 folder for the embedded file.**	

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Slide 172  **Note:** **ROI helps to justify the cost of reviews. It is calculated as follows:** **ROI = Net Savings / Detection Cost** **Where, Net Savings = Cost to Repair – Cost Avoidance** **Detection Cost = Cost of Preparation Effort + Cost of Effort to Conduct Review**	

Slide Content	Use this space for your notes
Slide 173  **Refer to the Supporting Documents_Day 1 for the embedded file.**	

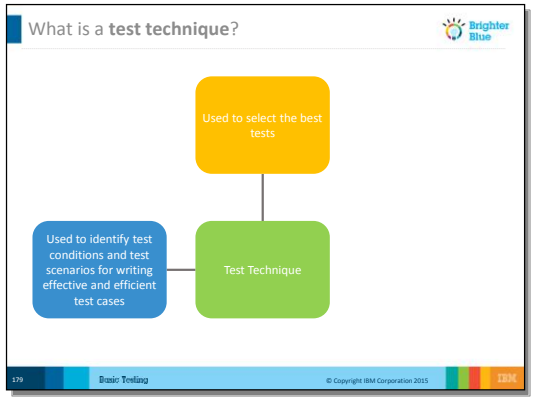
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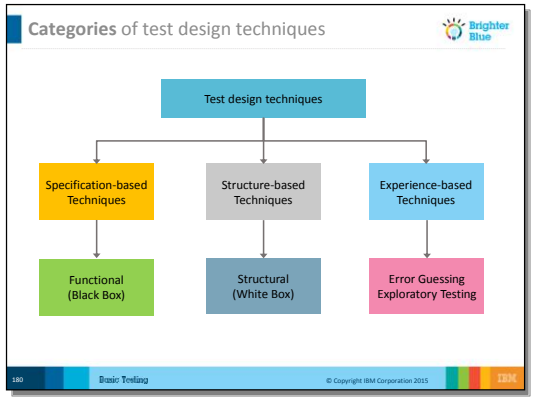
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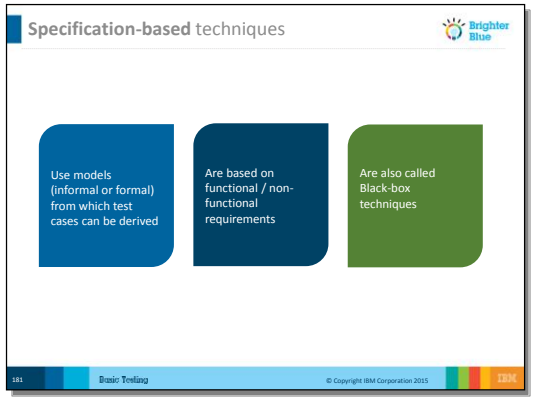
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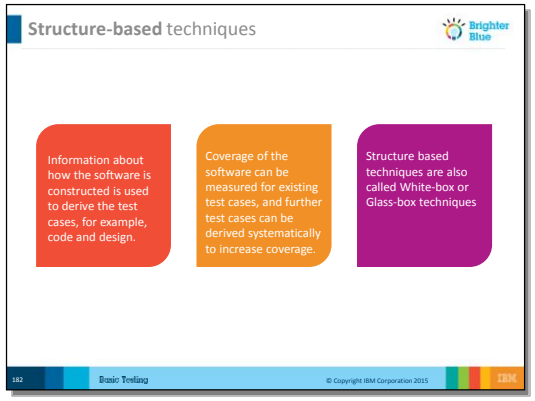
Module 4: Testing Design Techniques and Approach

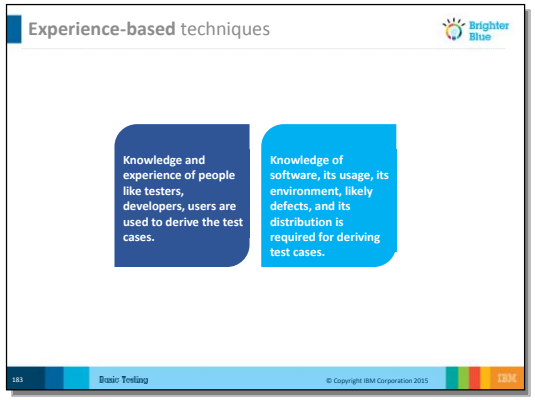
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Slide 177  **At the end of this module, you should be able to:** ▪ **Identify test conditions for designing test cases** ▪ **Describe categories and types of test design techniques** ▪ **List the components of a test case template**	

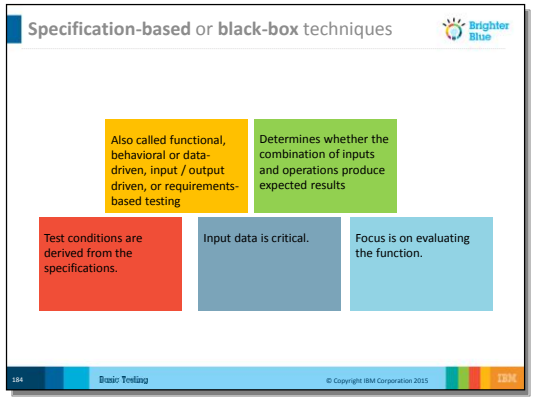
Slide Content	Use this space for your notes
Slide 178  ▪ **Test techniques are used to select the best tests from the infinite number of all possible tests for a given system.** ▪ **Test techniques are used to identify test conditions and test scenarios through which effective and efficient test cases can be written.** ▪ **Test techniques help to achieve high test coverage and define tests that will provide insight into the quality of the test object.**	

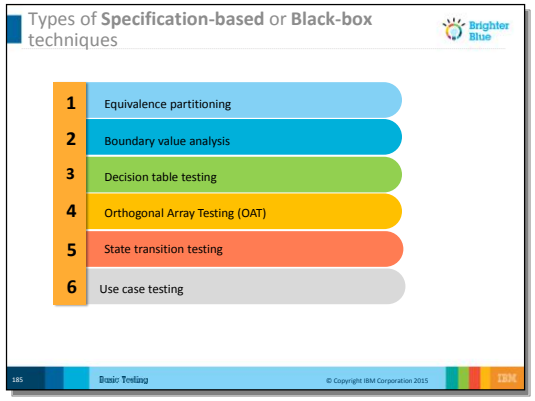
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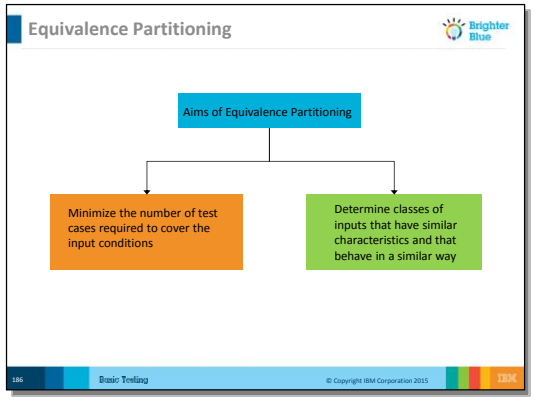
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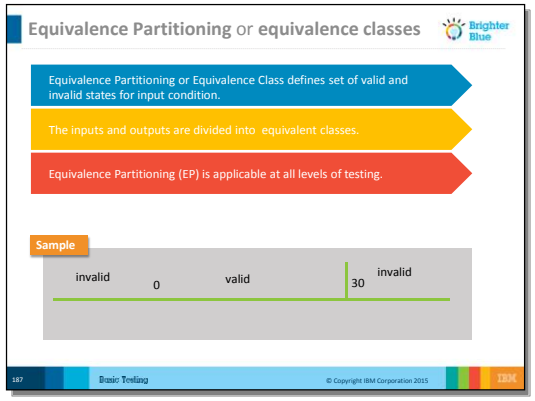
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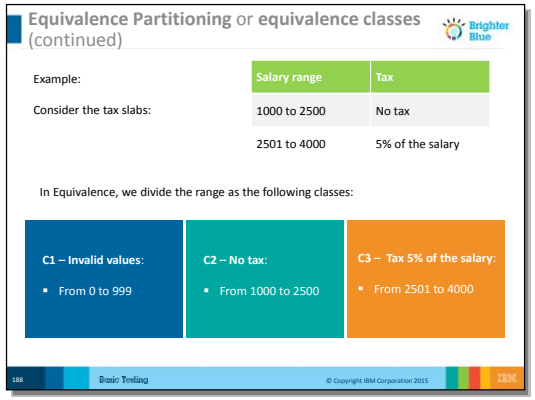
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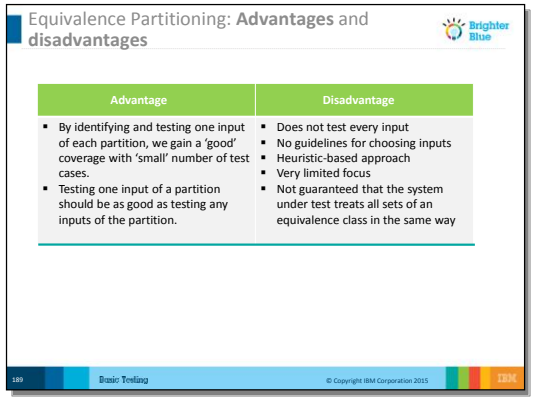
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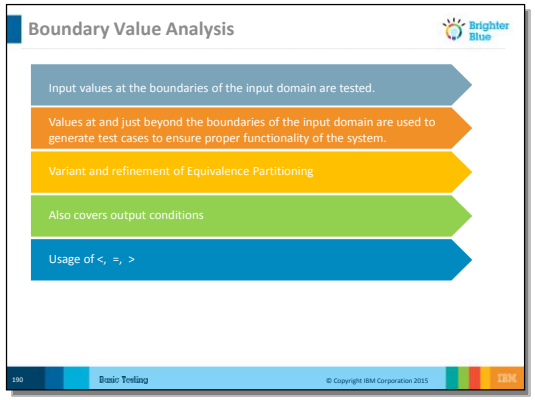
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Slide 185  **Aims:** ■ **Minimize the number of test cases required to cover the input conditions** ■ **Determine classes of inputs that have similar characteristics and that behave in a similar way. Two test cases based on inputs from the same Equivalence Class are likely to reveal same bug.** **Identifying Equivalence Classes (ECs):** ■ **If input specifies a RANGE of valid values, define one EC in the range and one outside each end.**	

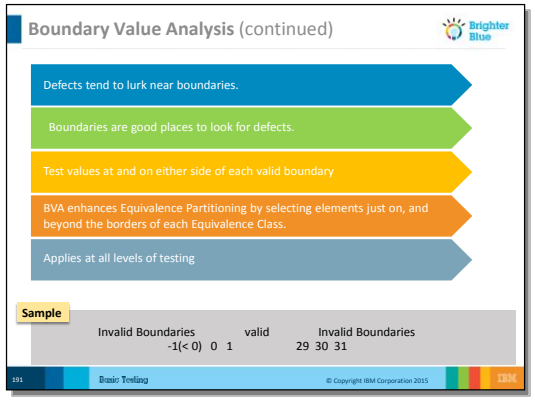
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▪ If input specifies a NUMBER (N) of valid values, define one valid EC and two invalid ECs (none, and more than N). ▪ If input specifies a SET of valid values, define one valid EC (within) and one invalid EC (not in). ▪ If there is a reason to believe the program handles each valid input differently, each valid input is an EC. ▪ If input specifies Must Be situation, define one valid EC (is) and one invalid EC (is not). ▪ If there is a reason to believe that elements in an EC are not handled in an identical manner, subdivide the EC into smaller ECs.	

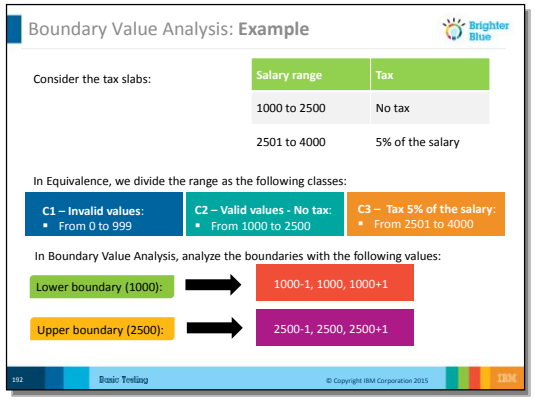
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Slide 187  **To put this in simpler words, since it is practically infeasible to do exhaustive testing, the next best alternative is to check whether the program extends similar behavior or treatment to a certain group of inputs. If such a group of values can be found in the input domain, treat them together as one equivalent class and test one representative from this class.**	

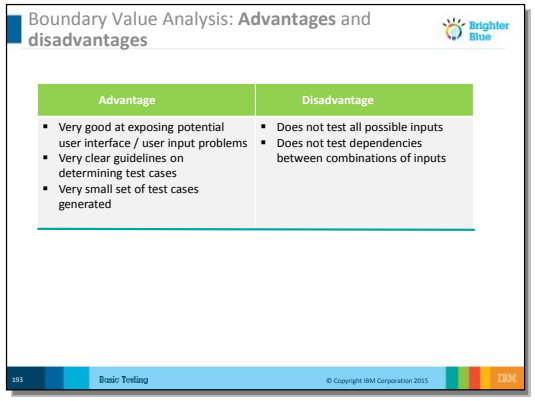
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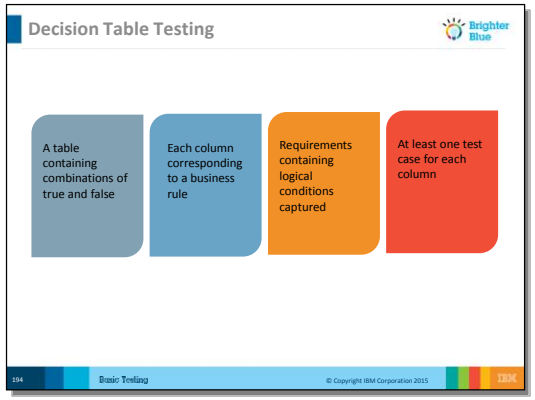
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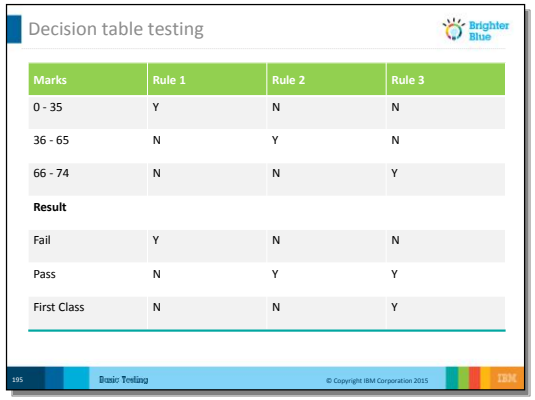
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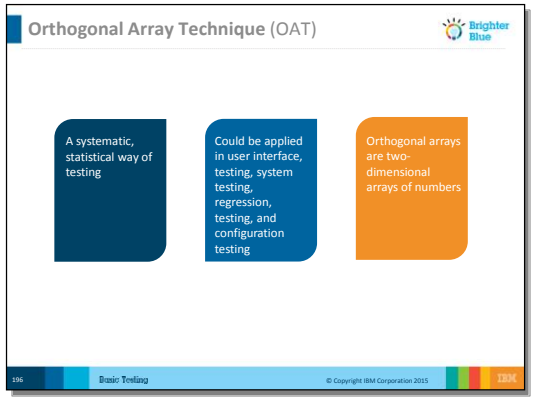
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Slide 191  **Consider that for salary between 1000-2500----->no tax and from 2501-4000----->tax 5% of their Salary.** **In EQUIVALENCE : we divide the range as the following classes:** **- from 0 to 999 is put together in one class c1- invalid values** **- from 1000 to 2500 in class c2 – no tax- Valid values** **- from 2501 to 4000 in class c3- 5% of their salary** **Boundary Value: taking the above example** **We would be analyzing the lower boundary (1000) with values as 1000-1,1000 and 1000+1** **Similarly the upper boundary (2500) would be tested with values as 2500-1, 2500 and 2500+1 for the class which is eligible for No Tax.**	

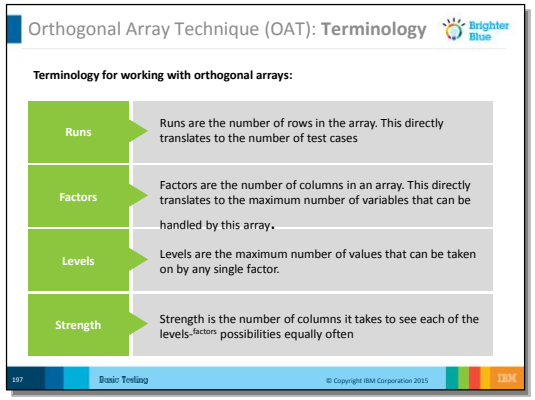
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Similarly we can also do the boundary value analysis for the class which is charged with 5% tax.	

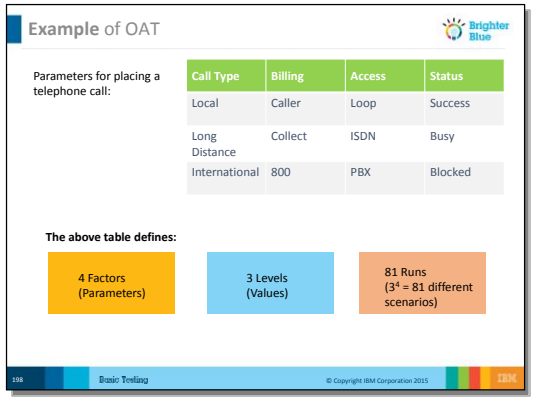
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Slide 193  **Characteristics of Decision Table Testing:** ▪ **A table contains combinations of true and false for all input conditions and the resulting actions for each.** ▪ **Each column corresponds to a business rule that defines a unique combination of conditions that result in the execution of the actions associated with that rule.** ▪ **Requirements that contain logical conditions can be captured.** ▪ **At least one test case is written per column.**	

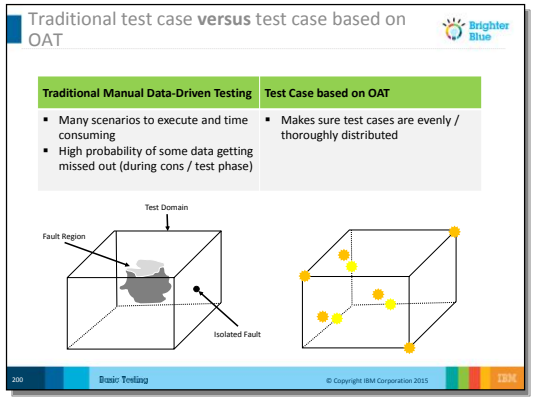
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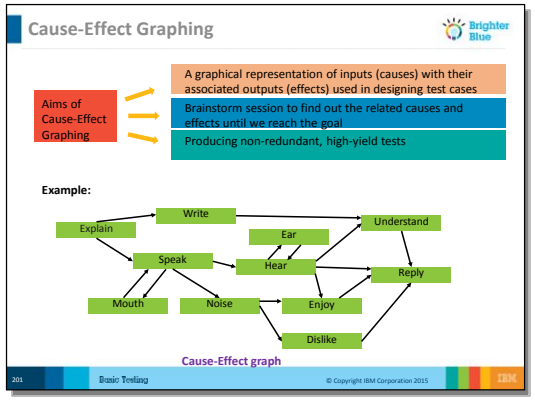
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Slide 195  **Characteristics of Orthogonal Array Technique:** ▪ **Orthogonal Array Technique / Testing is a systematic, statistical way of testing.** ▪ **Orthogonal arrays could be applied in user interface testing, system testing, regression testing, and configuration testing.** ▪ **Orthogonal arrays are two-dimensional arrays of numbers which possess the interesting quality that by choosing any two columns in the array you receive an even distribution of all the pair-wise combinations of values in the array.**	

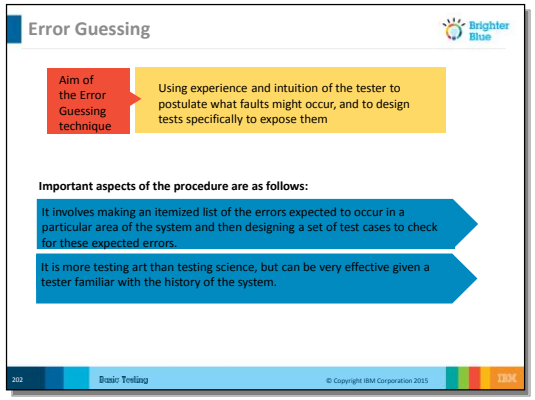
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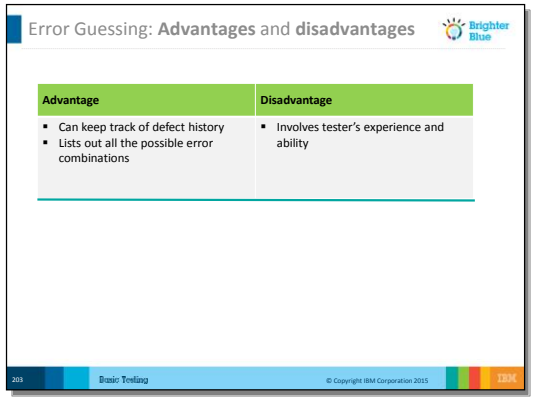
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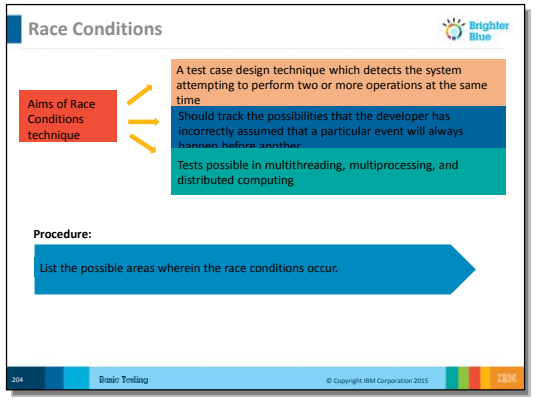
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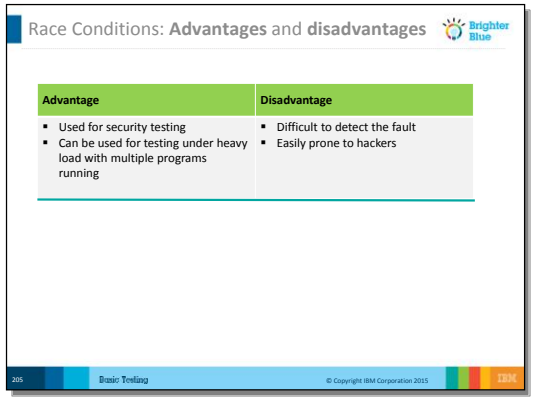
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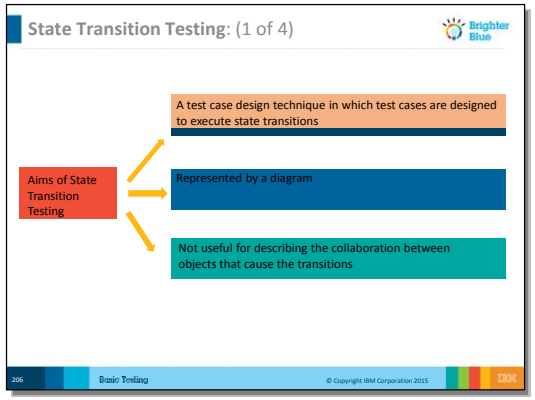
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Slide 201  **Aim:** ▪ **A test case design technique where the experience and intuition of the tester is used to postulate what faults might occur, and to design tests specifically to expose them** **Procedure:** ▪ **Involves making an itemized list of the errors expected to occur in a particular area of the system and then designing a set of test cases to check for these expected errors** ▪ **Empty or Null Strings / Lists**	

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▪ Zero Instances / Occurrences ▪ It is more testing art than testing science, but can be very effective given a tester familiar with the history of the system.	

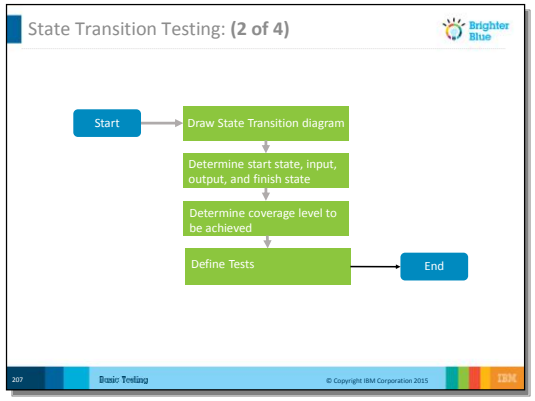
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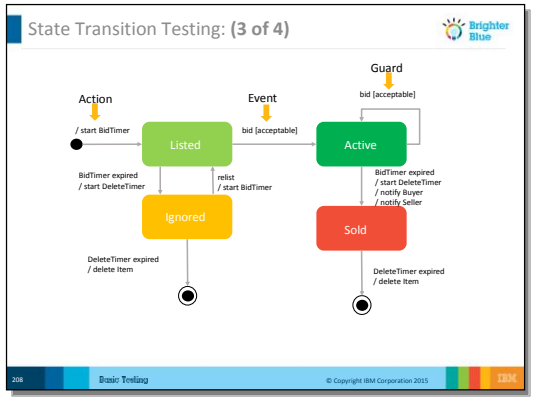
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Slide 203  **Procedure:** ▪ **List the possible areas wherein the race conditions occur:** ▪ **File open / close** ▪ **Variable checks** ▪ **Signals** ▪ **Access checks** ▪ **Locking**	

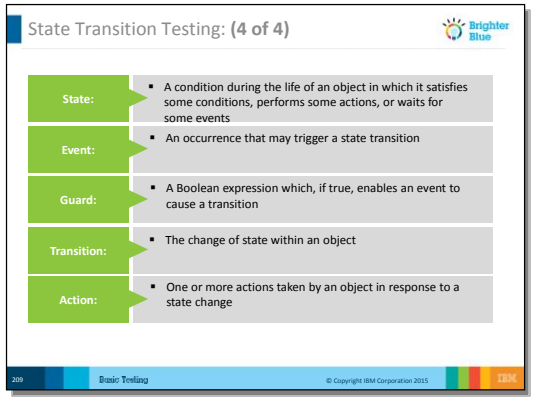
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Slide 205  **Aim:** ▪ **A test case design technique in which test cases are designed to execute state transitions** ▪ **Represented by a diagram describing:** ▪ **All of the states that an object can have** ▪ **The events under which an object changes state (transitions)** ▪ **The conditions that must be fulfilled before the transition will occur (guards)** ▪ **The activities undertaken during the life of an object (actions)**	

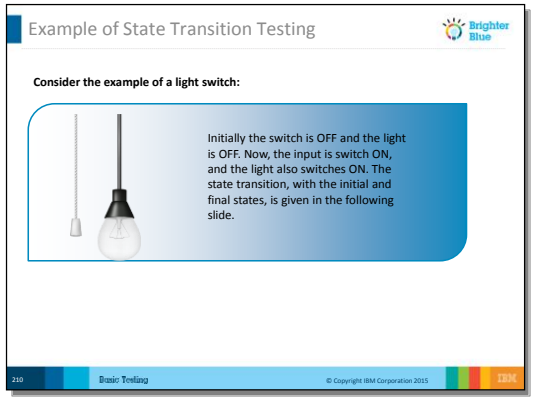
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▪ Not useful for describing the collaboration between objects that cause the transitions	

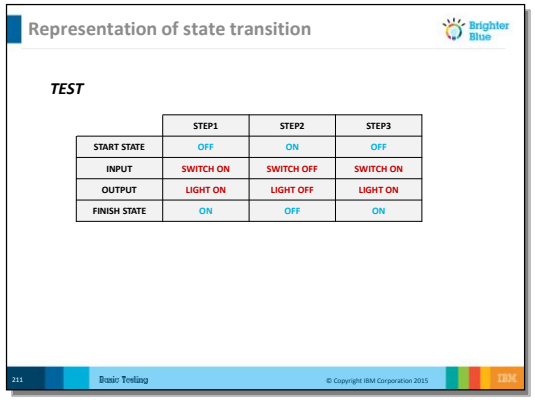
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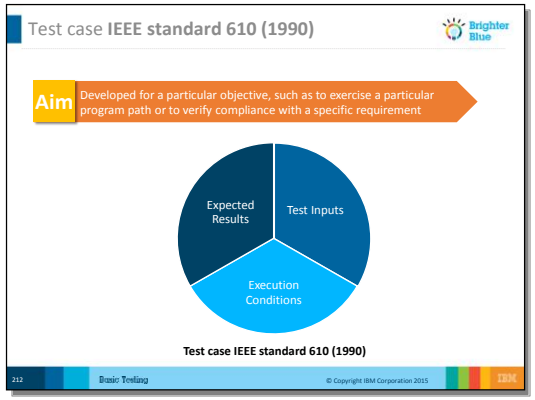
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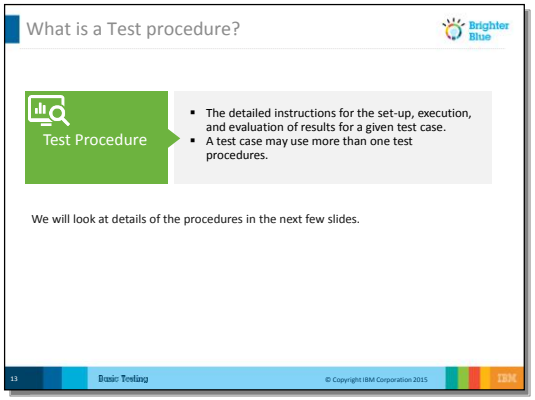
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Slide 208  Terminology: ▪ Notation: For those not familiar with the notation used for State Transition diagrams, some explanation is in order. ▪ State: A condition during the life of an object in which it satisfies some conditions, performs some actions, or waits for some events ▪ Event: An occurrence that may trigger a state transition. Event types include an explicit signal from outside the system, an invocation from inside the system, the passage of a designated period of time, or a designated condition becoming true.	

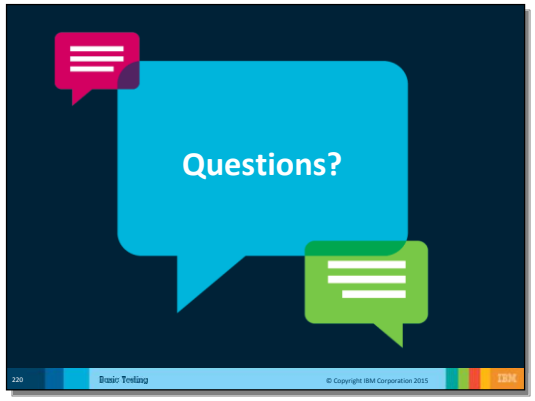
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▪ Guard: A Boolean expression which, if true, enables an event to cause a transition ▪ Transition: The change of state within an object ▪ Action: One or more actions taken by an object in response to a state change	

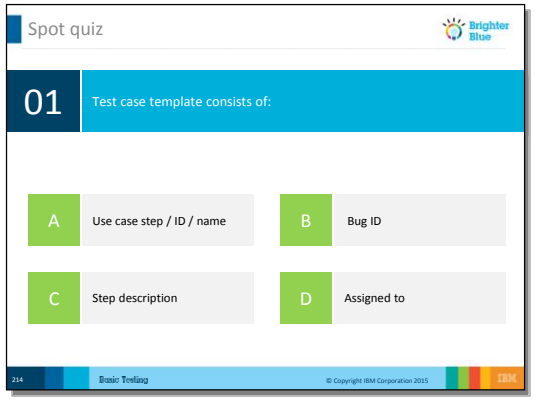
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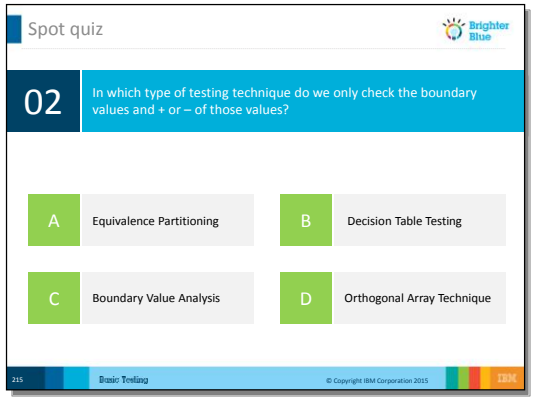
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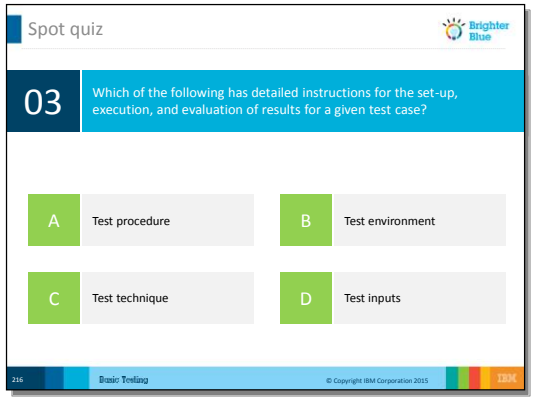
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Slide 211  **A set of:** ▪ **Test Inputs** ▪ **Execution Conditions** ▪ **Expected Results** **Developed for a particular objective, such as to exercise a particular program path or to verify compliance with a specific requirement**	

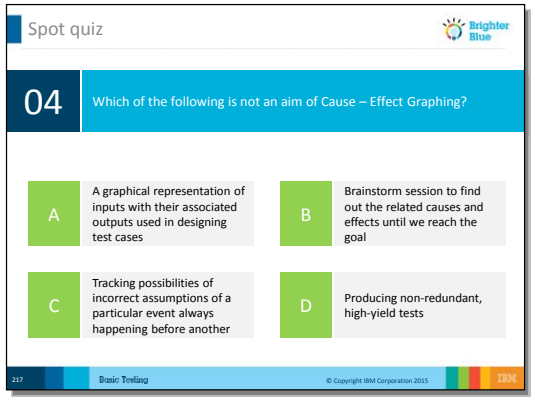
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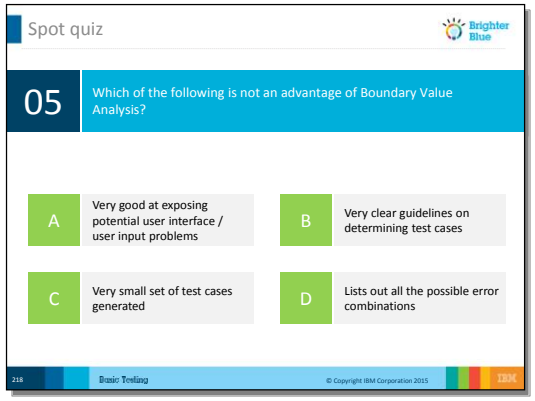
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Slide 213  **The key takeaways from this module are:** ▪ **Identifying test conditions for designing test cases** ▪ **Describing categories and types of test design techniques** ▪ **Listing the components of a test case template**	

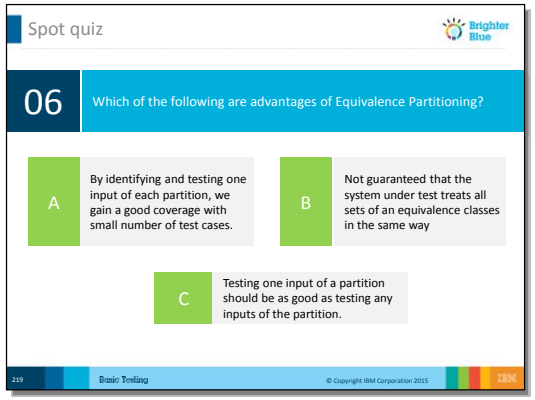
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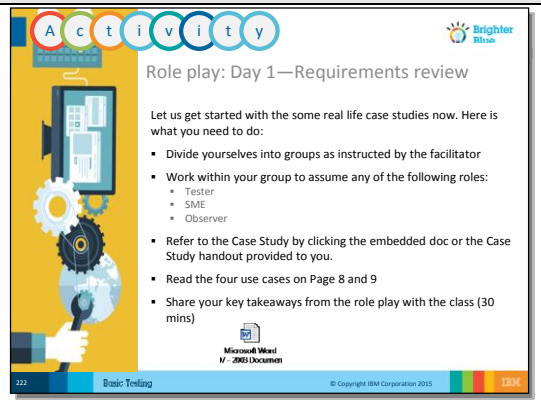
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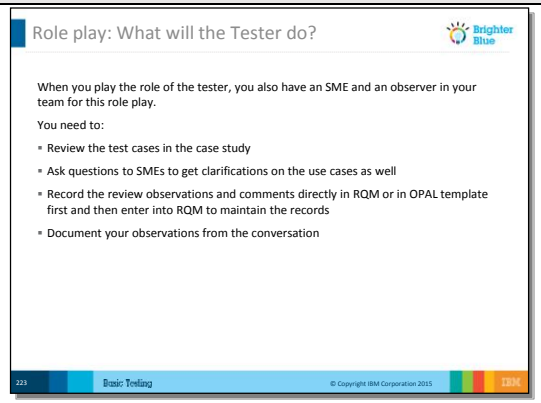
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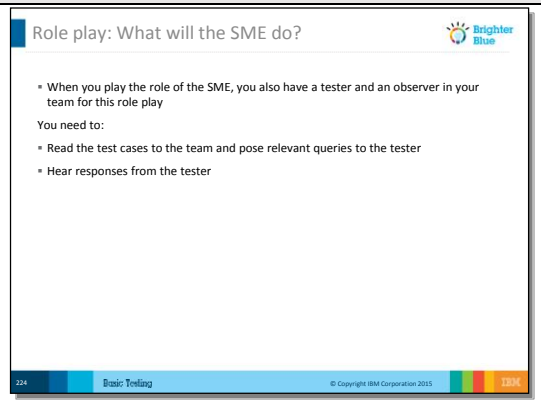
Module 5: Case Study

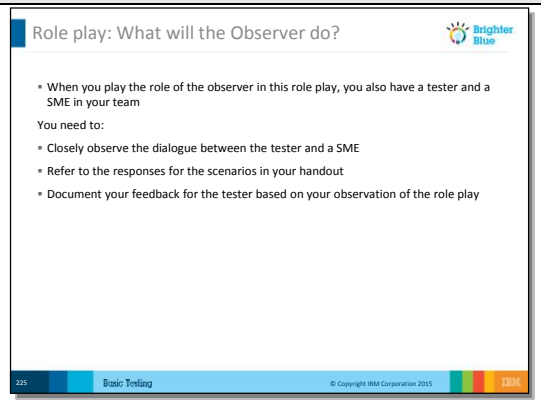
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 **At the end of this module, you should be able to:** • **Review the test cases in the case study** • **Ask questions to SMEs to get clarifications on the use cases** • **Record the review observations and comments directly in OPAL template first and then enter into RQM to maintain the records**	
Slide 221	

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 **Refer to the Requirements Review doc in the Supporting Documents_Day 1 folder for the embedded file.**	
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